TGT Technology Global-eSIM API_2.0_EN

1 OVERVIEW

1.1 Introduction

Welcome to TGT Global Cloud Communications Platform! We provide powerful, stable, and easy-to-integrate global eSIM card business APIs, dedicated to helping your business seamlessly integrate and manage global communication resources, enabling your enterprise to deliver instant, reliable, and cost-effective connectivity solutions.

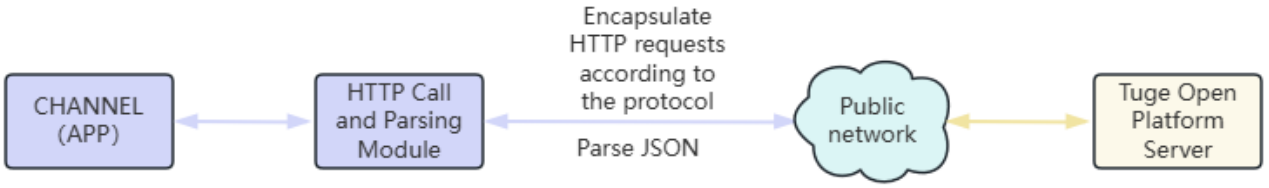
TGT Technology partners can seamlessly integrate our eSIM technology into your platform, empowering you to sell quickly and easily. Features include:Obtain Product Package, Create Order, product top-up/renewal, Order status and eSIM card installation status retrieval, etc.

1.2 Quick start

1.2.1 API Request Overview Diagram

TGT Global Cloud Communications Platform OpenAPI is based on HTTP protocol for invocation， Channels may initiate integration by encapsulating HTTP requests according to the platform's interface invocation specifications， The following primarily outlines instructions for conducting API integration testing through self-encapsulated HTTP requests.

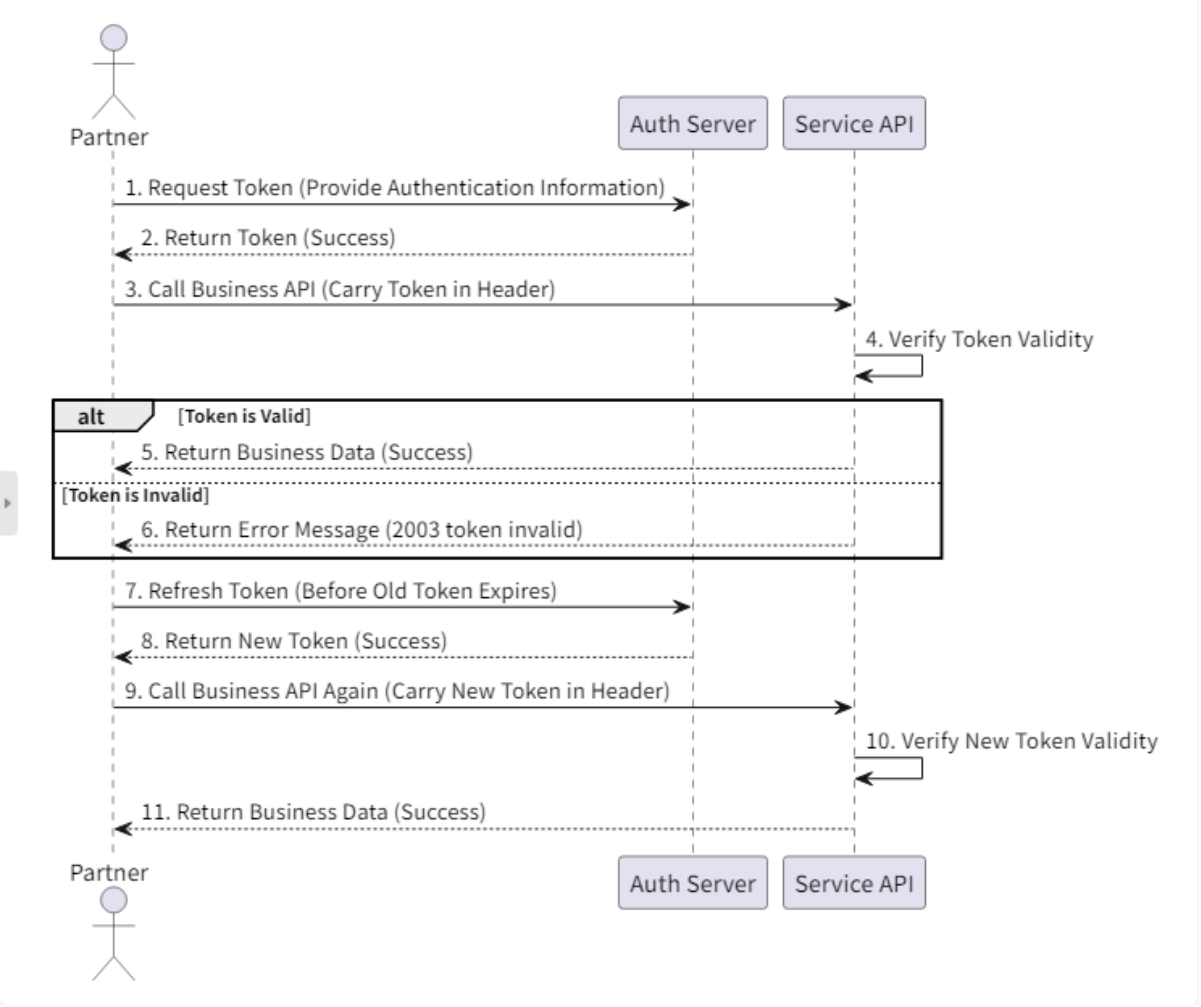
The API request process is as follows: Fill in parameters > Assemble HTTP request > Initiate HTTP request> Receive HTTP response > Parse JSON results. The main API request process is illustrated below:



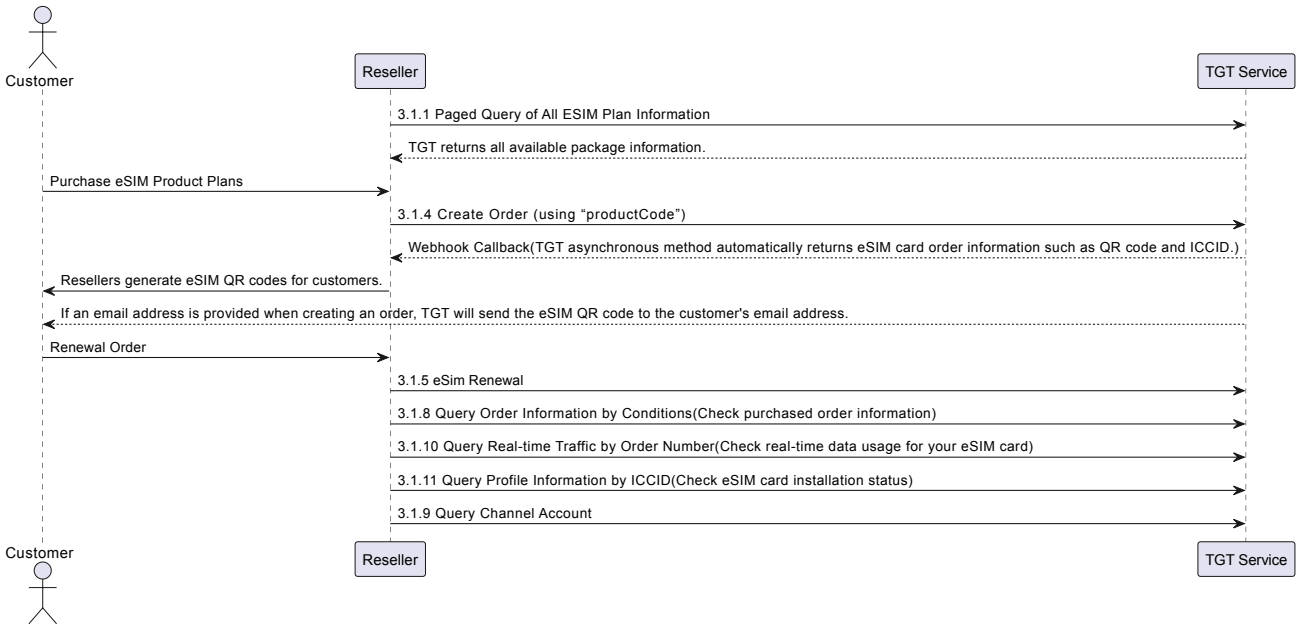
1.2.2 REST API overview

This endpoint allows you to obtain an access token, which is required to make requests to the TGT Partner API.Using your client account ID (accountId) and API secret (secret), you will receive a token valid for **24 hours.**

Important Notes: It is best to store the token in your API client and only request a new token when the current one expires or becomes invalid.



1.2.3 Primary API Interface Integration Process



1.3 Attribute descriptions

Attribute	Description
DAILY_PACK	Daily Data Plan, allows daily data allocation and resets on a specified cycle type.
DATA_PACK	Data Plan，One-Time Data Plan Usage within a specified period
AUTO_ACTIVATE	The package activates automatically. Within the order's validity period, you may activate and use it on any day. Once activated, the timer starts. The system will automatically update the actual activation period based on the usage start date, eliminating the need to specify a start time.
ACTIVATE_ON_ORDER	Specify an activation package and set a future start date during order placement to activate it. This is commonly used when users purchase in advance, with the plan becoming active on the designated date and the validity period commencing then.
usagePeriod	Active usage period: The actual period during which a package product remains valid and usable after activation.
validityPeriod	The validity period of the product package when not activated for use. Within this period, you may scan the eSIM QR code on any day to activate and install it.
periodType	Cycle Type 24H：The activation period lasts for 24 hours from start to finish. Calendar day：Activation start time to carrier card type specified time zone end time
top up	Top up，Add data allowance: When your original daily plan reaches its speed limit for the day, you can purchase a top-up package to resume usage. This does not extend the plan's validity period.A master order can correspond to multiple refueling package orders. (Note: Currently only supports C4 card type and daily package products)
Renewal	Renewal，Refers to purchasing a new Package on the same eSIM card to extend usage. This supports purchasing any product Package offered by this carrier, ensuring customers can continue using the eSIM card without needing to replace it.
accountId	Channel account ID required for API requests or logging into the TGT portal
secret	API key, the encryption information required for API requests

qrCode	The QR code information for the eSIM card profile is used by channel customers to generate their own eSIM card QR codes, and also serves to manually input the activation code for installation on mobile devices.
orderNo	TGT System eSIM Card Order Number
channelOrderNo	The channel system allows customization of order numbers based on specific usage scenarios.
Standard eSIM card	eSIM cards installed on mobile phones and other terminal devices by scanning the eSIM card's QR code or manually entering the activation code.
SMS Push eSIM Card	The local US eSIM card (US-T-Mobile) is activated via SMS push. The mobile device receives an activation message to install the eSIM card, If this type of eSIM card is not required, you may disregard it.
DAYPASS	Order Type: Daily Data Plan Place an order for designated activation （Daily Pack+Activate on Order）
DAILY	Order Type: Daily Data Plan Auto-Activate（Daily Pack+Auto-Activate）
MULTIPLEMONTHS	Order Type: Data Plan Place an order for designated activation（Data Pack+Activate on Order）
MULTIPLEMONTHS_AUTO	Order Type: Data Plan Auto-Activate（Data Pack+Auto-Activate）
productCode	The essential parameters for purchasing a product Package, and also the only non-modifiable parameters of the Product Package.
iccid	ICCID stands for Integrated Circuit Card Identifier, which is the unique identification code for each SIM card. It is a numeric code consisting of 19 or 20 digits.
imsi	IMSI is the number used by the network to uniquely identify a mobile subscriber's identity.
msisdn	msisdn is a mobile phone number.
QPS	Query Per Second，API interface query frequency per second
cardType	cardType，The TGT system creates operators card identifiers based on different operators cards to distinguish between different operators cards.

Response Time	Average time for interface response
Sandbox environment	Products configured for sandbox environments in channel system API testing are intended solely for API interface testing and cannot be used for actual operations. They do not involve order fees. Once API testing is successful, transition to the production environment.
Production Environment	TGT Commercial Sales Official Platform: Channel customers must first create an account in the production environment through business channels to use the platform. All products in the production environment are genuine and available for purchase, and creating orders will incur order fees.

2 User Notice

2.1 Establish business cooperation

Channel clients must establish a business partnership through TGT sales managers. API integration must be initiated by the business sales manager to proceed with API testing.

2.2 Obtain the test account ID and secret

TGT FAE engineers provide API documentation for Tuoge eSIM services, along with sandbox environment channel test accounts, passwords, and secret keys.

2.3 The channel provides a callback address.

- Callback mechanisms are used for asynchronous processing. Callbacks are also known as **webhook** and **notification**。The channel must provide a callback address in advance, which will be configured by TGT's FAE within the TGT system. The channel must implement idempotent design for this interface.
- "eventType"Fields are used to distinguish the categories of callback notification events. When the channel system receives a request notification from TGT, the response format must be:{"code":"0000","msg":"success"}

2.4 API test

- The TGT Cloud Platform supports user interface portal login (OSS Portal) and external API integration. During API testing, users can log in to the portal to review API test results.
- Users can integrate with API interfaces as needed based on usage scenarios, with no restrictions on client-side code formats (supporting Java/PHP/C/Python/JavaScript, etc.).

2.5 Production Environment

After the channel integration is completed in the test environment, the FAE engineer will verify it and request the provision of production environment account IDs and secret encryption parameters.

- Note: Creating orders in the production environment will incur order fees.

2.6 API QPS

- All API interfaces in this document have recommended QPS (**Queries Per Second**)request values. Strictly adhere to the specified limits outlined in the API documentation.
- The QPS per endpoint is the constraint value for all channel API requests. Ensure your implementation adheres to this limit to avoid rate limiting errors.

2.7 Important Notes

- All request and response data is encoded in UTF-8 format.
- Request Method: Uses the HTTP standard **POST** protocol.
- Set the Content-Type header in the request to application/json; charset=UTF-8.
- The request header must include a token.
- All times referenced in the interface are in **UTC+0** time zone.
- The time format in the interface is yyyy-mm-dd'T'HH:mm:ss'Z', for example: 2025-01-01T13:00:00Z

2.8 API List

API	Request path	Description
4.1 Pagination query for all eSIM product packages	/eSIMApi/v2/products/list	Retrieve information on all officially available package products under the channel account.
4.2 Single eSIM Product Package Inquiry	/eSIMApi/v2/products/detail	You can use the productCode to look up specific product package information.
4.3 Card Feature Inquiry	/eSIMApi/v2/card	Query card-related attribute information based on card type
4.4 Place Order（Standard eSIM）	/eSIMApi/v2/order/create	Create eSIM card order
4.5 eSim Renewal（Standard eSIM）	/eSIMApi/v2/order/renew	Renew your eSIM Package
4.6 Place Order（SMS Push eSIM）	/eSIMApi/v2/order/create/push	Create eSIM card order for push mode

4.7 Pagination Query for Order Information	/eSIMApi/v2/order/list	Retrieve information about orders that have been created
4.8Filter and query order information	/eSIMApi/v2/order/orders	Retrieve corresponding order information based on specific query conditions.
4.9 Check Account balance	/eSIMApi/v2/account/balance	Check the balance of the channel account
4.10 Real-time Traffic Data Query	/eSIMApi/v2/order/usage	Check real-time usage data for your eSIM card
4.11 eSIM Profile Inquiry	/eSIMApi/v2/iccid/profile	Query eSIM card profile installation and activation information
4.12 Termination Order Plan (Applies to C4 Cards Only)	/eSIMApi/v2/order/terminate	Discontinuation of Products Using Specific Card Types (C4)
4.13 Place top up order	/eSIMApi/v2/order/topup/create	Purchase a refill pack for a specific card type (C4)
4.14 Paginated Query of Top-up Orders	/eSIMApi/v2/order/topup/list	Check order details for purchased Refueling packages
4.15 Query Top-up Orders Associated with the orderNo	/eSIMApi/v2/order/topups	Retrieve Refueling package order information associated with the main order number

3 REST API

TGT Global Cloud Communications Platform enables you to integrate core connectivity features directly into your applications or systems.Use this API to empower your users to effortlessly view, purchase, and manage travel eSIMs, thereby enhancing your product offerings.

3.1 API Modes

To facilitate debugging and API integration, TGT partners can conduct testing within the TGT API sandbox environment. Upon successful API testing, TGT will provide production environment credentials, including

username, password, and secret key.

Mode	BaseUrl(HTTPS)	Description
Sandbox environment	https://enterpriseapisandbox.tugegroup.com:8070/openapi	Provided to customers for testing the development API interface environment
Production Environment	https://enterpriseapi.tugegroup.com:8070/openapi	Production environment, used for API integration with the client's actual sales platform. Transition to production is permitted only after successful API testing in the sandbox environment.

3.2 Public Attribute Description

Calling any API returns response parameters. Currently supported common parameters include:

Parameter Name	Type	Description
code	String	Error code returned when the request fails
msg	String	Error message returned when the request fails
subCode	String	Sub-error code returned upon request failure
subMsg	String	Sub-error message returned upon request failure
data	Object	Business Response Information

Note: For specific details regarding subcode and subMsg, refer to the code result set defined in each API. For specific details regarding data, refer to the data result set defined in each API.

3.3 Business Parameters

In addition to mandatory public parameters, API calls must also include any business-level parameters specific to the API. Refer to the API documentation for details on the business-level parameters for each API.

3.3.1 User Authentication Information

To prevent malicious tampering by hackers during API calls, a token must be included with every API request. The token is valid for 24 hours by default. Once expired, it must be refreshed; otherwise, the API interface will return a 2003 error indicating the token has expired.

3.3.2 Authorize to obtain a new token

Interface Description：The interface completes identity verification by passing in the accountId and secret, returning a new access token for subsequent authorized access to the interface.

BaseUrl：https://***/oauth/token

Request Parameters

Parameter Name	Type	Required	Example value	Description
accountId	String	true	TGT_Channel	Authorized Account
secret	String	true	jzXUuQVIlFwf3peM	Authorization Key, provided by TGT Administrator

Response Parameters

Parameter Name	Type	Example value	Description
code	String	0000	0000=sucess
msg	String	sucess	Information about error codes
data	Object		Token information, returns TokenInfo object
TokenInfo Object Field Description			
accessToken	String	756e816e8de648a9826e945f7e872bcf	Authorized Account
expires	long	86400	Expiration time, in seconds

3.3.3 Refresh Token

Interface Description：Before the token expires, this interface allows you to directly extend the token's validity period while keeping the token value unchanged. Users can continue accessing services with the original token without needing to log in again, ensuring uninterrupted access.

BaseUrl：https://***/oauth/refreshToken

Request Parameters

Parameter Name	Type	Required	Example Value	Description
accountId	String	true	TGT_Channel	Authorized Account
accessToken	String	true	756e816e8de648a9826e945f7e872bcf	Authorization token

Response Parameters

Parameter Name	Type	Example value	Description
code	String	0000	0000=sucess
msg	String	sucess	Information about error codes
data	Object		Token information, returns TokenInfo object
TokenInfo Object Field Description			
accessToken	String	756e816e8de648a9826e945f7e872bcf	Authorization token
expires	long	86400	Expiration time, in seconds

3.3.4 API Request Header Configuration

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```
//Settings header
HttpHeaders headers = new HttpHeaders();
headers.set(HttpHeaders.CONTENT_TYPE, "application/json;charset=UTF-8");
headers.set("Authorization", "Bearer " + accessToken);
```

3.3.5 API Request Example

Taking the retrieval of eSIM plan information as an example, the specific steps are as follows:

(1) Set parameter values

Business Parameters:

- pageSize = 100
- page = 1

- lang = "en"

(2) Assembling HTTP Requests

Note: URL-encode all parameter names and values using UTF-8 (parameter order is flexible, but the signature parameter must be included), then initiate the request via POST.

POST request:

复制代码

```
  Timestamp t = new Timestamp(System.currentTimeMillis());
  JSONObject params = new JSONObject();
  params.put("pageSize", 10);
  params.put("page ", 1);
  params.put("lang ", "en");

  URI uri;
  HttpEntity<JSONObject> request;
  uri = new URIBuilder(BaseUrl+"/eSIMApi/v2/products/list").build();

  //Settings header & Parameters
  HttpHeaders headers = new HttpHeaders();
  headers.set(HttpHeaders.CONTENT_TYPE, "application/json;charset=UTF-8");
  headers.set("Authorization", "Bearer " + token);
  request = new HttpEntity<JSONObject>(params, headers);

  // Executed by the client (sending) a POST request
  ResponseEntity<JSONObject> response = restTemplate.postForEntity(uri,
request, JSONObject.class);
  if (null != response && null != response.getBody()) {
      System.out.println(response.getBody());
  } else {
      System.out.println(response.getStatusCode());
  }
```

4 eSIM API

4.1 Pagination query for all eSIM product packages

Business Description

Paginate queries for all product packages and pricing information under channel accounts. Queries can also be filtered by product package type. The interface returns only product packages available for official sale. This endpoint helps you synchronize eSIM product packages with your systems.

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/products/list

QPS Limit：30

Response Time：800ms

4.1.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
pageNum	int	true	1	Page number
pageSize	int	true	100	Page size (max 100)
productType	String	false	DAILY_PACK	Product type: DAILY_PACK (//Daily Data Plan) DATA_PACK （//data plan）
cardType	String	false	C4	Types of eSIM Cards from Different Operators
usagePeriod	Integer	false	4	Actual usable time after eSIM activation
periodType	Integer	false	0	Data reset time or end time 0：24hours 1：Calendar day
lang	String	false	en	Language (English: en, Japanese: ja-JP, Chinese: zh-CN), default English

4.1.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description

data	Object		Data encapsulation entity, returns an **EsimProductRes** object
EsimProductRes Object Field Description			
total	Integer	100	Total data count
list	JSONArray		Data encapsulation entity, returns array of **ProductInfo** objects
ProductInfo Object Field Description			
productCode	String	E-111-SC-AU-eO1-T-15D/60D-5G	Product ID: This ID is required when placing orders via the API to purchase subscription packages.
productName	String	【eSIM】Europe 32 Countries 5G/15 Days (60-Day Validity) (eO1)	Package Product Name Supports internationalized display. Based on the value of the input parameter “lang” (language parameter, e.g., zh-CN = Chinese, en = English), it automatically matches and returns the product name in the corresponding language.
productType	String	DAILY_PACK	Product Type: DAILY_PACK： Daily Data Plan（Daypass） DATA_PACK： Data Plan（Data package）
countryCodeList	List<String>	["US","CA"]	National Code Set, ISO 3166 Country Codes
mccList	List<String>	["310","311"]	National MCC Collection
netPrice	Decimal(28,2)	10.0	Settlement Price/Wholesale Net Price
periodType	Integer	0	Cycle Type 0：24hours 1：Calendar day
usagePeriod	Integer	0	Activation Period (the actual period during which the product can be used) Based on the type of periodType,

			0：24hours 1：Calendar day
validityPeriod	Integer	60	Order validity period: The period during which the product package but unused. Within this period, you may scan the eSIM QR code on any day to activate and install the plan.
dataLimited	String	N	Traffic Limit Indicator (Y, N): Y indicates limited traffic, N indicates unlimited traffic. When set to Y, the total traffic volume must be retrieved via the dataTotal field.
dataTotal	Integer	1	Total Data Volume
dataUnit	String	GB	Data Units (GB, MB, KB)
lastModifiedTime	String	2025-08-27T09:00:00Z	Last Modified Date
activeType	String	AUTO_ACTIVATE	Activation Method: AUTO_ACTIVATE：Auto-Activate ACTIVATE_ON_ORDER：Specified Activation
ruleDesc	String	1GB high-speed data, valid until used up	Package Rules Description The value of the input parameter “lang” (language parameter, e.g., zh-CN = Chinese, en = English) will be used.
operatorDesc	String	ATT	Esim Carrier Description
apnDesc	String	CMHK	APN description. If this field is not empty, it must be returned.
highSpeed	String	500M	High-speed value
limitSpeed	String	128Kbps	Speed limit value
showGradeContent	String	N(K)10240(M)2000(K)	Speed rating
cardType	String	C2	Card Type

topupInfoList	JsonArray		Top-up Package Information List, an array of **TopupInfo** objects
TopupInfo Object Field Description			
topupId	String	1925504013246783489	Refueling Package ID
topupName	String	500MB add-on	Refueling Package Name
topupSize	String	500	Data value
topupUnit	String	MB	Data Units (GB, MB)
topupPrice	Decimal(28,2)	1.00	Top-up Package Price

Request Example 复制代码

```
{
  "pageSize":100,
  "lang":"en",
  "pageNum":1
    }
```

Response Example 复制代码

```
{
 "code":"0000",
 "msg":"success",
 "data":
     {"total":2,
    "list":[{
        "productCode":"A-002-ES-AU-T-30D/180D-3GB(A)",
        "productName":"【ESIM】Israel 3GB/30 Days (180-Day Validity) (M1) (A)",
        "productType":"DATA_PACK",
        "countryCodeList":["IL"],
        "mccList":["425"],
        "netPrice":1.10,
        "periodType":0,
        "usagePeriod":30,
        "validityPeriod":180,
        "dataLimited":"Y","dataTotal":3,
        "dataUnit":"GB",
```

```
        "lastModifiedTime":"2025-11-06T10:54:10Z",
        "activeType":"AUTO_ACTIVATE",
        "operatorDesc":"Annatel",
        "apnDesc":"M1-annatel",
        "cardType":"M1"},
        {"productCode":"A-136-ES-AU-C4-1D/60D-1GB",
         "productName":"【Esim】Asia 5 Countries 1 Day (1GB High-Speed/Day)(C4)
(AU)",
        "productType":"DAILY_PACK",
        "countryCodeList":["JP","CN","SG","KR","MY"],
        "mccList":["440","441","460","525","450","502"],
        "netPrice":2.00,
        "periodType":0,
        "usagePeriod":1,
        "validityPeriod":60,
        "dataLimited":"N",
        "lastModifiedTime":"2025-11-06T11:20:57Z",
        "activeType":"AUTO_ACTIVATE",
        "ruleDesc":"1GB high-speed data per day, down to 384kbps",
        "operatorDesc":"NTT Docomo/China Mobile/Singtel/StarHub/M1",
        "apnDesc":"C4",
        "highSpeed":"1GB",
        "showGradeContent":"1GB/384kbps",
        "cardType":"C4",
        "topupInfoList":[{
                "topupId":"22",
                "topupName":"1GB add-on",
                "topupSize":"1",
                "topupUnit":"GB",
                "topupPrice":1.00}]
        }}
```


4.2 Single eSIM Product package Inquiry

Business Description

Query eSIM package and pricing information under the channel account based on “productCode”. The API only returns currently available data package. Currently, eSIM services only include daily packages, data packages, and their corresponding top-up packages.

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/products/detail

QPS Limit：50

Response Time：500ms

4.2.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
productCode	String	true	A-002-ES-AU-C4-D-2D/4D-Unlimited	Product Code
lang	String	false	en	Language (English: en, Japanese: ja-JP, Chinese: zh-CN), default English

4.2.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description
data	Object		Data encapsulation entity, returns **ProductInfo** object
ProductInfo Object Field Description			
productCode	String	E-111-SC-AU-eO1-T-15D/60D-5G	Product ID: This ID is required when placing orders via the API to purchase subscription packages.
productName	String	【eSIM】Europe 32 Countries 5G/15 Days (60-Day Validity) (eO1)	Package Product Name Supports internationalized display. Based on the value of the input parameter “lang” (language parameter, e.g., zh-CN = Chinese, en = English), it automatically matches and returns the product name in the corresponding language.

productType	String	DAILY_PACK	Product Type: DAILY_PACK： Daily Data Plan DATA_PACK： Data Plan
countryCodeList	List<String>	["US","CA"]	National Code Set, ISO 3166 Country Codes
mccList	List<String>	["310","311"]	National MCC Collection
netPrice	Decimal(28,2)	10.0	Settlement Price/Wholesale Net Price
periodType	Integer	0	Cycle Type 0：24hours 1：Calendar day
usagePeriod	Integer	4	Activation Period (the actual period during which the product can be used) Based on the type of periodType, 0：24hours 1：Calendar day
validityPeriod	Integer	60	Order validity period: The period during which the product package but unused. Within this period, you may scan the eSIM QR code on any day to activate and install the plan.
dataLimited	String	N	Traffic Limit Indicator (Y, N): Y indicates limited traffic, N indicates unlimited traffic. When set to Y, the total traffic volume must be retrieved via the dataTotal field.
dataTotal	Integer	1	Total Data Volume
dataUnit	String	GB	Data Units (GB, MB, KB)
lastModifiedTime	String	2025-08-27T09:00:00Z	Last Modified Date

activeType	String	AUTO_ACTIVATE	Activation Method: AUTO_ACTIVATE：Auto-Activate ACTIVATE_ON_ORDER：Specified Activation
ruleDesc	String	1GB high-speed data, valid until used up	Package Rules Description The value of the input parameter “lang” (language parameter, e.g., zh-CN = Chinese, en = English) will be used.
operatorDesc	String	ATT	Esim Carrier Description
apnDesc	String	CMHK	APN description. If this field is not empty, it must be returned.
highSpeed	String	500M	High-speed value
limitSpeed	String	128Kbps	Speed limit value
showGradeContent	String	N(K)10240(M)2000(K)	Speed rating
cardType	String	C2	Card Type
topupInfoList	JsonArray		Top-up Package Information List, an array of **TopupInfo** objects
TopupInfo Object Field Description			
topupId	String	1925504013246783489	Refueling Package ID
topupName	String	500MB add-on	Refueling Package Name
topupSize	String	500	Data value
topupUnit	String	MB	Data Units (GB, MB)
topupPrice	Decimal(28,2)	1.00	Top-up Package Price

Error Code Explanation

subCode（Detailed Return Code）	**subMsg**（Return Code Description）	Solution
4001	The product does not exist.	The product does not exist
4002	The product is not on sale.	The product has been discontinued
4000	Other Anomalies	Contact the TGT administrator

请求示例 复制代码

```
{
  "productCode":"A-002-ES-AU-T-30D/180D-3GB(A)",
  "lang":"en"
}
```

Response Example 复制代码

```
{
  "code":"0000",
  "msg":"success",
    "data":{
         "productCode":"A-002-ES-AU-T-30D/180D-3GB(A)",
          "productName":"【ESIM】Israel 3GB/30 Days (180-Day Validity) (M1)
(A)",
          "productType":"DATA_PACK","countryCodeList":["IL"],
          "mccList":["425"],
          "netPrice":1.10,
          "periodType":0,
          "usagePeriod":30,
          "validityPeriod":180,
          "dataLimited":"Y",
          "dataTotal":3,
          "dataUnit":"GB",
          "lastModifiedTime":"2025-11-06T10:54:10Z",
          "activeType":"AUTO_ACTIVATE",
          "operatorDesc":"Annatel",
          "apnDesc":"M1-annatel",
          "cardType":"M1"}}
```

4.3 Card Feature Inquiry

Business Description

Query the card's specific feature based on its type to determine whether it supports renewal, real-time data queries, and the maximum number of active orders simultaneously.

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/card

QPS Limit：50

Response Time：200ms

4.3.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
cardType	String	true	C4	Card Type

4.3.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description
data	Object		Data encapsulation entity, returns **ProductInfo** object
CardTypeInfo Object Field Description			
cardType	String	C2	Card Type
timeZone	String	UTC+8	Time Zone (Natural Day Time Zone)
renewFlag	Boolean	true	Renewal Indicator: true:Support, false:Not supported

supportGetUsage	Boolean	true	Does support real-time data queries true:Support, false:Not supported
renewCount	Integer	1	Maximum number of active orders at any given time

Error Code Explanation

subCode（Detailed Return Code）	**subMsg**（Return Code Description）	Solution
6003	The card type does not exist.	The card type does not exist. Please verify the correct card type.

Request Example 复制代码

```
{"cardType":"C4"}
```

响应示例 复制代码

```
{
  "code":"0000",
  "msg":"success",
    "data":{
      "cardType":"C4",
      "renewFlag":true,
      "supportGetUsage":false,
      "renewCount":99}}
```

4.4 Place Order（Standard eSIM）

Business Description

This interface enables channels to create individual orders for standard eSIMs. Special attention is required:TGT System employs an asynchronous callback model for order result notifications via this interface, rather than synchronous returns. The specific notification logic is detailed in the following description.

Core Interaction Logic（Asynchronous Notification Mechanism）

- **Callback Address Configuration:**The channel must provide the callback address for receiving actual order results in advance. This address is pre-configured into the system by a TGT specialist. After an order is placed, the TGT system will send the actual order details to the channel via the callback address. The channel then processes the business logic based on the received notification.
- **Basic Response and Asynchronous Notification Triggering:**
  - After the channel initiates an order request, it will first receive an order number returned by the TGT system (serving as the unique identifier for the order);
  - Once the eSIM card order process is completed, the TGT system will automatically invoke the pre-configured callback address. It will push the complete eSIM card information via asynchronous notifications (refer to the “eSIM Activation Callback Interface Documentation” for specific notification parameters).
- **Key Recommendations (Preventing Repeated Callbacks):**To prevent duplicate receipt of the same card activation result notification within the channel system, it is recommended that the callback interface `callback` under development， Add idempotent processing logic based on order numbers (e.g., determine whether the notification has been processed by checking the order number; if already processed, return success directly without re-executing the business process).

Exception Handling Recommendations (Retry Mechanism)

To enhance the stability of API calls, it is recommended that the channel system implement a request retry mechanism with the following rules:

- **Unique Identifier Configuration:**Each time an order request is initiated, the following must be included in the request parameters: `idempotencyKey` field(Customized by the channel (must be a globally unique string (UUID v4 is the preferred choice));
- **Retry Trigger Scenarios:**When exceptions occur such as request interruptions, response timeouts, or unknown system errors, and the channel has not confirmed receipt of a valid response from the TGT system, a retry may be triggered.
- **Key Rules for Retries:**When initiating a retry request again, you must pass in the exact same `idempotencyKey` as the previous request to prevent multiple invalid orders from being created due to duplicate requests.

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/order/create

Response Time：500ms

4.4.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description

productCode	String	true	E-111-SC-AU-eO1-T-15D/60D-5G	Product Code
startDate	String	false	2025-10-06	Order Start Date (1) If the product is automatically activated（activeType=**AUTO_ACTIVATE**）Package, the start time field does not require a value to be passed; it defaults to the current order time. Passing a value will not take effect. (2) If the product is activated on a specified date（activeType=**ACTIVATE-ON-ORDER**）, If the current field is not passed, it represents the current order time. If a time is passed, it must be in GMT+0 format. This time field must not be earlier than the current GMT+0 time.
email	String	false	123@email.com	Email, If you need to send a QR code to the cardholder, please enter their email address(es). Multiple email addresses should be separated by commas, with a maximum length of 200 characters.
channelOrderNo	String	ture	12345	Channel Order Number (Customizable order number for channel customers) (Maximum length: 100 characters)
idempotencyKey	String	ture	550e8400-e29b-41d4-a716-446655440000)	The maximum length of the request ID must not exceed 64 characters. In exceptional cases, such as request interruptions or timeouts, if the channel does not receive a response from the TGT system, please pass the previous idempotencyKey when retrying. It must be a globally unique string (UUID v4 is the preferred choice).

4.4.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description
data	Object		Data encapsulation entity, returns an **OrderInfo** object
OrderInfo Object Field Description			
orderNo	String	SE20250513170929410001	Order Number

Error Code Explanation

subCode（Detailed Return Code）	**subMsg**（Return Code Description）	Solution
5013	Channel currency does not exist!	Channel currency does not exist. Contact TGT business personnel to configure channel currency.
4001	The product does not exist.	Product does not exist. Create new product and list it.
5000	The order is still being processed.	Order is being processed. Reuse idempotent value for placing orders during the order number generation phase.

请求示例 复制代码

```
{
   "productCode":"A-002-ES-AU-T-30D/180D-3GB(A)",
    "idempotencyKey":"12356",
    "channelOrderNo":"1234578"
}
```

Response Example 复制代码

```
{"code":"0000",
  "msg":"success",
  "data":{
  "orderNo":"SE20251121190119280001"}
}
```

4.5 eSim Renewal（Standard eSIM）

Business Description

Renew data plans for eSIM cards to repurchase new data packages when the eSIM card runs low on data.Not all cards support renewal; this must be confirmed with the business department beforehand.The interface for “Query card properties by card type” can be used to determine renewal eligibility via the renewFlag field. (Commercially provided quotation sheets will also indicate whether renewals are supported.)

Core Interaction Logic（Asynchronous Notification Mechanism）

- **Callback Address Configuration:**The channel must provide the callback address for receiving actual order results in advance. This address is pre-configured into the system by a TGT specialist. After an order is placed, the TGT system will send the actual order details to the channel via the callback address. The channel then processes the business logic based on the received notification.
- **Basic Response and Asynchronous Notification Triggering:**
  - After the channel initiates an order request, it will first receive an order number returned by the TGT system (serving as the unique identifier for the order);
  - Once the eSIM card order process is completed, the TGT system will automatically invoke the pre-configured callback address. It will push the complete eSIM card information via asynchronous notifications (refer to the “eSIM Activation Callback Interface Documentation” for specific notification parameters).
- **Key Recommendations (Preventing Repeated Callbacks):**To prevent duplicate receipt of the same card activation result notification within the channel system, it is recommended that the callback interface `callback` under development，Add idempotent processing logic based on order numbers (e.g., determine whether the notification has been processed by checking the order number; if already processed, return success directly without re-executing the business process).

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/order/renew

Response Time：500ms

4.5.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
productCode	String	true	A-002-ES-AU-C4-D-2D/4D-Unlimited	Product Code
startDate	String	false	2025-10-06	Order Start Date (1) If the product is automatically activated（activeType=**AUTO_ACTIVATE**）Package，the start time field does not require a value to be passed; it defaults to the current order time. Passing a value will not take effect. (2) If the product is activated on a specified date（activeType=**ACTIVATE-ON-ORDER**），If the current field is not passed, it represents the current order time. If a time is passed, it must be in GMT+0 format. This time field must not be earlier than the current GMT+0 time.
email	String	false	123@email.com	Email，If you need to send a QR code to the cardholder, please enter their email address(es). Multiple email addresses should be separated by commas, with a maximum length of 200 characters.
channelOrderNo	String	true	12345	Channel Order Number (Customizable order number for channel customers) (Maximum length: 100 characters)
iccid	String	true	12345	ICCID

idempotencyKey	String	true	550e8400-e29b-41d4-a716-446655440000	The maximum length of the request ID must not exceed 64 characters. In exceptional cases, such as request interruptions or timeouts, if the channel does not receive a response from the TGT system, please pass the previous idempotencyKey when retrying. It must be a globally unique string (UUID v4 is the preferred choice).

4.5.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description
data	Object		Data encapsulation entity, returns **ProductInfo** object
OrderInfo Object Field Description			
orderNo	String	SE20250513170929410001	orderNo

Error Code Explanation

subCode（Detailed Return Code）	**subMsg**（Return Code Description）	Solution
5013	Channel currency does not exist！	Channel currency does not exist The channel must be configured with a currency that meets system requirements.

4001	The product does not exist.	No product information found. Please try a different product code.
4003	ESIM only supports monthly user package (month), daily package (day), and multiplemonths products.	Supports only Daily Packages, Monthly Packages, and Data Packages.
4004	ESIM products only support ESIM device model group products	Product model grouping is not supported.
4005	ESIM product configuration information is illegal (supplier is empty)!	The supplier value for the eSIM product is empty.（suppliers）
4006	ESIM product configuration information is illegal (carrier is empty)!	The card type value for the eSIM product is empty.（carrier）
4007	The product belongs to the incorrect platform	Default:CLOUDSIMA (The product's platform is incorrect)
4008	The channels of this business model do not support calling this interface	The channels for this business model do not support calling this interface. Recommendation: Change the business model
4009	This channel is not configured for the product price	Configure Product Pricing
4010	Channel account balance is insufficient, please top up	Insufficient funds
4011	Package settlement price is illegal!	The package settlement price is illegal! Modify settlementprice
4012	The order start date cannot be less than the current time	订单开始时间不能早于当前时间
4013	product is unvisible,can not to add order	channelVisible Change to 1
5040	The latest renewal time has passed, the renewal is not supported!	The renewal deadline has passed. Renewal is no longer supported!

5041	The number of renewals has exceeded the limit, do not support renewal!	Cancel individual orders
5042	This type of card renewal is not supported!	The card type has renewal rules configured but does not support renewal. A different card type is required.
5043	Hello, the renewal package you have selected does not match your current card, please select a matching package to renew your subscription. For further assistance, please contact our Customer Service!	Changing packages: The selected renewal plan does not match your current card. Please choose a compatible plan for renewal.
5047	Carrier cards are not configured for renewal!	The TGT system is not configured for renewal.
5048	Order status not satisfied, no continuing fee!	Order status not satisfied, no continuing fee!
5049	The device does not belong to the channel. Select a new one	Replace the card number belonging to the channel

Request Example 复制代码

```
{
  "iccid":"89852342714026530002",
  "productCode":"E-01-F2-ES-AU-T-7D/60D-1GB(A)",
  "idempotencyKey":"145632",
  "channelOrderNo":"88963589"
}
```

响应示例 复制代码

```
{
  "code":"0000",
  "msg":"success",
  "data":{
      "orderNo":"SE20251121191732344 15"}
}
```

4.6 Place Order（SMS Push eSIM）

Business Description

This interface enables channels to create individual orders using the US-TMOBILE push model for local U.S. cards.Special Note: The TGT system uses an asynchronous callback model for order result notifications via this interface, rather than synchronous returns. See the following description for specific notification logic.

Core Interaction Logic（Asynchronous Notification Mechanism）

- **Callback Address Configuration:**The channel must provide the callback address for receiving actual order results in advance. This address is pre-configured into the system by a TGT specialist. After an order is placed, the TGT system will send the actual order details to the channel via the callback address. The channel then processes the business logic based on the received notification.
- **Basic Response and Asynchronous Notification Triggering:**
  - After the channel initiates an order request, it will first receive an order number returned by the TGT system (serving as the unique identifier for the order);
  - Once the eSIM card order process is completed, the TGT system will automatically invoke the pre-configured callback address. It will push the complete eSIM card information via asynchronous notifications (refer to the “eSIM Activation Callback Interface Documentation” for specific notification parameters).
- **Key Recommendations (Preventing Repeated Callbacks):**To prevent duplicate receipt of the same card activation result notification within the channel system, it is recommended that the callback interface `callback` under development，Add idempotent processing logic based on order numbers (e.g., determine whether the notification has been processed by checking the order number; if already processed, return success directly without re-executing the business process).

Ordering Rules

- **Scope of Application:**This section applies only to eSIM products using the “push mode,” such as the “US-TMOBILE” card (whether integration with this mode is required must first be confirmed with Tuoge Business; if integration is not needed, this section may be skipped).
- Special Requirements:
  1. **Activation Time Configuration:**The request must specify the activation start date for the eSIM card, which must be set to “current UTC time + 6 days” (e.g., if the current UTC time is 2024-05-01, the activation start date must be set to 2024-05-07).

Exception Handling Recommendations (Retry Mechanism)

To enhance the stability of API calls, it is recommended that the channel system implement a request retry mechanism with the following rules:

- **Unique Identifier Configuration:**Each time an order request is initiated, the following must be included in the request parameters: `idempotencyKey` field(Customized by the channel (must be a globally unique string (UUID v4 is the preferred choice));
- **Retry Trigger Scenarios:**When exceptions occur such as request interruptions, response timeouts, or unknown system errors, and the channel has not confirmed receipt of a valid response from the TGT system, a retry may be triggered.
- **Key Rules for Retries:**When initiating a retry request again, you must pass in the exact same `idempotencyKey` as the previous request to prevent multiple invalid orders from being created due to duplicate requests.

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/order/create/push

Response Time：500ms

4.6.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
productCode	String	true	E-111-SC-AU-eO1-T-15D/60D-5G	Product Code
startDate	String	true	2025-10-06	Order Start Date Currently, this card type only supports productCode with specified activation (interface 3.1 returns activeType=ACTIVE_ON_ORDER) and requires the specified activation start date to be the current date plus 6 days. This is a mandatory field.
channelOrderNo	String	true	29b-41d4-a716-446655440001	Channel Order Number (Customizable order number for channel customers) (Maximum length: 100 characters)
eid	String	true	20123456789000000000111111111118	The unique identifier for an eSIM, composed of a 32-digit number.
imei2	String	true	867120029852612	Composed of 15 to 17 digits or (digits + letters)

idempotencyKey	String	true	550e8400-e29b-41d4-a716-446655440000	The maximum length of the request ID must not exceed 64 characters. In exceptional cases, such as request interruptions or timeouts, if the channel does not receive a response from the TGT system, please pass the previous idempotencyKey when retrying. It must be a globally unique string (UUID v4 is the preferred choice).

4.6.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description
data	Object		Data encapsulation entity, returns **ProductInfo** object
OrderInfo Object Field Description			
orderNo	String	SE20250513170929410001	orderNo

Error Code Explanation

subCode（Detailed Return Code）	**subMsg**（Return Code Description）	Solution
5006	The IMEI2 must be 15-17 digits long!	Enter a field that meets the format requirements; IMEI2 length does not meet requirements (15-17 digits).

5007	The specified start date of the order cannot be less than the specified time of the package (the time two days after zero hour in the U.S. time zone)	Time must meet the time defined by the requirements.
5012	EID must be 32 characters, Numbers combinations!	Time must meet the time defined by the requirements.
5013	Channel currency does not exist!	Currency query error. Contact customer service.
4001	The product does not exist.	Add Product

Request Example 复制代码

```
{
  "productCode": "E-111-SC-US-TMOBILE-30D/60D-10G",  // Required, Push Mode
Exclusive Product Code
  "startDate": "2025-10-06",                       // Required, Activation
Start Date
                                                      (Current UTC + 6 days,
only supported by ACTIVATE_ON_ORDER)
  "channelOrderNo": "CHAN_PUSH20251006001",      // Required: Channel Custom
Order Number (Maximum 100 characters)
  "eid": "20123456789000000000111111111118",         // Required, eSIM Unique
Identifier (32-digit number)
  "imei2": "867120029852612",                      // Required: Device IMEI2
(15-17 digit/letter combination)
  "idempotencyKey": "550e8400-e29b-41d4-a716-446655440001"  // Required,
globally unique request ID
}
```

Response Example 复制代码

```
{
  "code": "0000",
  "msg": "success",
  "subCode": "",
  "subMsg": "",
  "data": {
    "orderNo": "PU20251006112233840003" }          // Push Mode Order Number
}
```

4.7 Pagination Query for Order Information

Business Description

Query information for placed orders based on pagination criteria, including eSIM card QR codes, order validity periods, order statuses, order numbers, ICCIDs, and other details.

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/order/list

QPS Limit：80

Response Time：200ms

4.7.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
pageNum	int	true	1	Page Number Lookup
pageSize	int	true	100	Items per page, maximum 100
orderStatus	String	false	NOTACTIVE	Order Status (NOTACTIVE: Not Activated, ACTIVATED: Activated, INUSE: In Use, USED: Used Up, EXPIRED: Expired, ABANDON: Abandoned, TERMINATION: Terminated)
orderTypeList	List<String>	false	DAYPASS	Order Type Collection（DAYPASS、DAILY、MULTIPLEMONTHS_AUTO、MULTIPLEMONTHS）
createdStartTime	String	false	2025-08-01T00:00:00Z	Start date (yyyy-MM-dd format), queries records created after the start date. If a value is provided, the end date is also required.
createdEndTime	String	false	2025-08-01T00:00:00Z	End time (yyyy-MM-dd format), query records where the creation time is less than or equal to the end time. If a value is passed, the start time is also required.

lang	String	false	en	Language (English: en, Japanese: ja-JP, Chinese: zh-CN), default English

4.7.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	String	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description
data	Object		Data encapsulation entity, returns **ProductInfo** object
OrderRes Object Field Description			
total	Integer	100	Data total
list	JSONArray		Data encapsulation entity, returns an array of **OrderInfo** objects
OrderInfo Object Field Description			
orderNo	String	TG202509251310149158	orderNo
productCode	String	D2309141025035378683	Product code
productName	String	np-cmlink-C3(AU日用包4GB/5天)	Product Name Supports internationalized display. Based on the value of the input parameter “lang” (language parameter, e.g., zh-CN = Chinese, en = English), it automatically matches and returns the product name in the corresponding language.
activatedStartTime	String	2025-08-01T00:00:00Z	Activation Start use Date
activatedEndTime	String	2025-08-05T23:59:59Z	Activation Expiration use Date

latestActivationTime	String	2025-09-24T21:10:36Z	Latest activation time
renewExpirationTime	String	2025-09-24T21:10:36Z	Latest renewal date
createdTime	String	2025-09-24T21:10:36Z	Order Creation Time
orderStatus	String	ABANDON	Order Status: NOTACTIVE: Not activated ACTIVATED: Activated INUSE: In use USED: Used up EXPIRED: Expired ABANDON: Abandoned TERMINATION: Terminated
profileStatus	String	unvaliable	eSIM Card Status “Inactive”,“unavailable” ‘Activated’,“activated” “Deleted”,“deleted” ‘Downloading’,“downloaded” (activation unsuccessful) "Disabled","disabled" "Not downloaded","nodownload" "Download failed","downloadfail" "Generation failed","failed" "Not generated","ungenerated"
qrCode	String	LPA:1$tgt.toprsp.com$B21B96B156C040B8A42233455FRDSADV	eSIM QR code information, in the format ofLPA:1$ *RSP Server Address* $ *Activation Code*
channelOrderNo	String	2342342342	Channel order number, a customizable order number for channel customers
orderType	String	DAYPASS	Order Type (DAYPASS: Specified Activation Daily Package, DAILY: Auto-Activation Daily Package, MULTIPLEMONTHS_AUTO: Auto-Activation Data Package, MULTIPLEMONTHS: Specified Activation Data Package)
cardInfo	Object		Return the **CardInfo** object

CardInfo Object Field Description			
iccid	String	95514350628283049898	ICCID
imsi	String	454120051751432	IMSI
msisdn	String	1670173	Mobile phone number
rentalContractNumber	String	8180508970	Lease Code

Error Code Explanation

subCode（Detailed Return Code）	**subMsg**（Return Code Description）	Solution
5037	Both createdStartTime and createdEndTime are either empty or not empty at the same time	createdStartTime and createdEndTime Either both are null, or both are not null.

Request Example 复制代码

```
{
  "pageSize":5,
  "pageNum":1
}
```

Response Example 复制代码


```
{
  "code":"0000",
    "msg":"success",
    "data":{"total":2,
            "list":[{"orderNo":"SE2025112119173234415",
            "productCode":"E-01-F2-ES-AU-T-7D/60D-1GB(A)",
            "productName":"【ESIM】UK 1GB/7 days (valid for 60 days) (F2)",
             "activatedStartTime":"2025-11-21T11:17:33Z",
            "activatedEndTime":"2025-11-28T11:17:32Z",
            "latestActivationTime":"2026-01-20T11:17:32Z",
            "renewExpirationTime":"2026-01-27T11:17:32Z",
            "createdTime":"2025-11-21T11:17:33Z",
            "orderStatus":"NOTACTIVE",
            "profileStatus":"activated",
```

```
"qrCode":"LPA:1$esiminfra.toprsp.com$98F57097621E451F8649135AC0A03011",
            "channelOrderNo":"88963589",
            "orderType":"MULTIPLEMONTHS_AUTO",
            "cardInfo":{"iccid":"89852342714026530002",
            "imsi":"454179386020002",
              "msisdn":"852136416970002"}},
            {"orderNo":"TG202511131717024170",
              "productCode":"E-184-ES-AU-eO1-D-10D/60D-10GB",
              "productName":"【Esim】Europe (11 Countries) 10 Days (10GB High-
Speed Data/Day) (eO1) (AU)",
              "activatedStartTime":"2025-11-13T09:17:22Z",
              "activatedEndTime":"2025-11-23T09:17:21Z",
              "latestActivationTime":"2026-01-12T09:17:21Z",
              "renewExpirationTime":"2026-01-22T09:17:21Z",
              "createdTime":"2025-11-13T09:17:22Z",
              "orderStatus":"NOTACTIVE",
              "profileStatus":"nodownload",

"qrCode":"LPA:1$esiminfra.toprsp.com$D0C23496B79C4179A7FA57E757C12299",
              "orderType":"DAILY",
              "cardInfo":{"iccid":"89852345127626040002",
              "imsi":"453128365970002",
              "msisdn":"852657436910002"}}]}}
```


4.8 Filter and query order information

Business Description

Retrieve order details by order number, channel order, or ICCID. At least one of these three criteria must be provided. Information includes the eSIM card's QR code, order validity period, order status, order number, ICCID, and other relevant details.

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/order/orders

QPS Limit：50

Response Time：200ms

4.8.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
orderNo	String	false	TG239829483838762	orderNo
iccid	String	false	89124893927372882317	ICCID
channelOrderNo	String	false	ORDER123	Channel Order Number, a customizable order number for channel customers
lang	String	false	en	Language (English: en, Japanese: ja-JP, Chinese: zh-CN), default English

4.8.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description
data	Object		Data encapsulation entity, returns an **OrderList** object
OrderList Object Field Description			
list	JSONArray		Data encapsulation entity, returns an array of **OrderInfo** objects
OrderInfo Object Field Description			
orderNo	String	TG202509251310149158	orderNo

productCode	String	ptsk-6	Product code
productName	String	SKTeSIM-10days_V	Product Name Supports internationalized display. Based on the value of the input parameter “lang” (language parameter, e.g., zh-CN = Chinese, en = English), it automatically matches and returns the product name in the corresponding language.
activatedStartTime	String	2025-08-01T00:00:00Z	Activation Start Use Date
activatedEndTime	String	2025-08-05T23:59:59Z	Activation Expiration Use Date
latestActivationTime	String	2025-08-05T23:59:59Z	Latest activation time
renewExpirationTime	String	2025-08-05T23:59:59Z	Latest renewal date
createdTime	String	2025-08-05T23:59:59Z	Ordering time
orderStatus	String	NOTACTIVE	Order Status: NOTACTIVE: Not activated ACTIVATED: Activated INUSE: In use USED: Used up EXPIRED: Expired ABANDON: Abandoned TERMINATION: Terminated
profileStatus	String	nodownload	eSIM Card Status “Inactive”,“unavailable” ‘Activated’,“activated” “Deleted”,“deleted” ‘Downloading’,“downloaded” (activation unsuccessful) "Disabled","disabled" "Not downloaded","nodownload" "Download failed","downloadfail" "Generation failed","failed" "Not generated","ungenerated"

qrCode	String	LPA:1$tgt.toprsp.com$B21B96B156C040B8A42233455FRDSADV	eSIM QR code information, in the format of LPA:1$ *RSP Server Address* $ *Activation Code*
channelOrderNo	String	1400819766678787878 9	Channel order number, a customizable order number for channel customers
orderType	String	DAYPASS	Order Type (DAYPASS: Specified Activation Daily Package, DAILY: Auto-Activation Daily Package, MULTIPLEMONTHS_AUTO: Auto-Activation Data Package, MULTIPLEMONTHS: Specified Activation Data Package)
CardInfo Object Field Description			
iccid	String	8982052207013909758F	ICCID
imsi	String	078945294185657	IMSI
msisdn	String	12345464	Mobile phone number
rentalContractNumber	String	8180508970	Lease Code

Error Code Explanation

subCode（Detailed Return Code）	**subMsg**（Return Code Description）	Solution
5032	orderNo, iccid, channelOrderNo, at least one is not empty!	Order Number, ICCID, Channel Order Number—at least one of these fields cannot be left blank!

Request Example 复制代码

```
{
  "iccid":"89852342714026530002",
  "orderNo":"SE2025112119173234415"
  }
```

Response Example 复制代码


```
{
  "code":"0000",
    "msg":"success",
    "data":{
          "list":[{"orderNo":"SE2025112119173234415",
            "productCode":"E-01-F2-ES-AU-T-7D/60D-1GB(A)",
            "productName":"【ESIM】UK 1GB/7 days (valid for 60 days) (F2)",
            "activatedStartTime":"2025-11-21T11:17:33Z",
            "activatedEndTime":"2025-11-28T11:17:32Z",
            "latestActivationTime":"2026-01-20T11:17:32Z",
            "renewExpirationTime":"2026-01-27T11:17:32Z",
            "createdTime":"2025-11-21T11:17:33Z",
            "orderStatus":"NOTACTIVE",
            "profileStatus":"activated",

"qrCode":"LPA:1$esiminfra.toprsp.com$98F57097621E451F8649135AC0A03011",
            "channelOrderNo":"88963589",
            "orderType":"MULTIPLEMONTHS_AUTO",
            "cardInfo":{"iccid":"89852342714026530002",
            "imsi":"454179386020002",
            "msisdn":"852136416970002"}}]}}
```


4.9 Check Account balance

Business Description

This endpoint can be used to query channel account and balance information.

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/account/balance

QPS Limit：50

Response Time：50ms

4.9.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
type	String	false	BASIC	Account Type: BASIC（Basic Account） DEPOSIT(Deposit Account)

4.9.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description
data	Object		Data encapsulation entity, returns an **AccountInfo** object
AccountInfo Object Field Description			
currency	String	USD	Settlement currency for channel accounts
accountId	String	tgt_salesapp	Channel coding
name	String	途鸽app	Channel Name
settlementType	String	CASH	Payment Methods（Pay as you go CASH、Monthly settlement MONTHSETTLE）
accountList	JSONArray		Data encapsulation entity, returns a **BalanceInfo** object
BalanceInfoInternal Field Description			
id	String	1	id

type	String	BASIC	Account Type（BASIC：Basic Account DEPOSIT：Margin Account）
status	String	ENABLE	Account Status（ENABLE：Enable, DISABLE：Not enabled）
balance	BigDecimal	1000.00	Account balance, default to two decimal places

Request Example 复制代码

```
{"type":"BASIC"}
```

Response Example 复制代码

```
{
  "code":"0000",
    "msg":"success",
    "data":{"currency":"USD",
    "accountId":"api-test-01-US",
    "name":"api-test一级渠道（美元现付）",
    "settlementType":"CASH",
    "accountList":[
            {"id":"27db3ec3fb1647319229e3497a5a7c37",
            "type":"BASIC",
            "status":"ENABLE",
            "balance":"99954.30"}]}}
```

4.10 Real-time Traffic Data Query

Business Description

Check real-time data information by order number

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/order/usage

QPS Limit：50

Response Time：800ms

4.10.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
orderNo	String	true	SE20250513170929410001	orderNo

4.10.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description
data	Object		Data encapsulation entity, returns a **UsageInfo** object
UsageInfo Object Field Description			
dataTotal	String	1000	Total data for the cycle, in MB Daily package, No such parameter returned
dataUsage	String	100	Current cycle data usage, in MB
dataResidual	String	900	Remaining data allowance for the current billing cycle, in MB Daily package, No such parameter returned
refuelingTotal	String	0	Daily Refueling package data total, in MB
qtaconsumption	String	100	Daily high-speed data usage, in MB

Error Code Explanation

subCode（Detailed Return Code）	subMsg（Return Code Description）	Solution
5002	The order does not exist.	Check if the order number exists
5005	The order card type does not support querying real-time traffic	Order card type does not support real-time data queries.
4068	Real-time traffic calculation in progress, please try again later!	In real-time data calculations, orders in a state where queries are not supported, or queries may fail due to exceptions.

Request Example 复制代码

```
{
  "orderNo": "SE20250513170929410001"  // Required, Order Number
}
```

Response Example 复制代码

```
{
  "code": "0000",
  "msg": "success",
  "data": {                // UsageInfo Object
    "dataTotal": "1000",   // Total Cycle Data（Unit: MB, This parameter is
not available for daily package.）
    "dataUsage": "100",    // Current Data Usage（Unit: MB）
    "dataResidual": "900", // Remaining data（Unit: MB, This parameter is not
available for daily package.）
    "refuelingTotal": "0", // Daily Refueling package data total（Unit: MB）
    "qtaconsumption": "100"  // Daily high-speed data usage（Unit: MB）
  }
}
```

4.11 eSIM Profile Inquiry

Business Description

Query profile information and renewal information based on ICCID.

Note:

1. There is a certain delay in eSIM order status.
2. Cancelled orders are not supported for query.

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/iccid/profile

QPS Limit：50

Response Time：200ms

4.11.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
iccid	String	true	89372000010000030001	ICCID

4.11.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description
data	object		Data encapsulation entity, returns a **ProfileInfo** object
ProfileInfo Object Field Description			
iccid	String	8997250212345555894F	ICCID
eid	String	89049032007408882600198987512345	eID, a globally unique 32-digit code, the unique identifier for an eSIM (embedded SIM card)

imsi	String	260036699995661	IMSI
state	String	Enabled	Esim profile state Downloaded (Downloading) Installed (Download completed) Error (Download retry limit exceeded) Unavailable (Revoked) Disabled Enabled Deleted Released (Initial state)
installDevice	String	iPhone 16 Pro	Installation device
installCount	String	1	Installation count
installTime	String	2025-11-10T14:27:53Z	Installation time
updateTime	String	2025-11-10T14:27:54Z	Update time

Error Code Explanation

subCode	subMsg	Solution
5004	Orders under the iccid do not exist or all have been cancelled.	Channel order does not exist or has been cancelled.
6002	The card profile file does not exist.	Profile information does not exist.

Request Example 复制代码

```
{"iccid":"89852342714026530002"}
```

Response Example 复制代码

```
{"code":"0000",
  "msg":"success",
  "data":{"iccid":"89852342714026530002",
  "imsi":"454179386020002",
  "state":"Enabled",
  "installCount":"0",
  "updateTime":"2025-11-13T10:22:11Z"}
```

```
}
```

4.12 Termination Order Plan (Applies to C4 Cards Only)

Business Description

For specific card types (**C4**), terminate the plan based on order number and card number. Not all cards support plan termination; confirmation with the business team is required. Note: Only activated and used plans can be terminated.

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/order/terminate

QPS Limit：50

Response Time：500ms

4.12.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
orderNo	String	true	TG2025112103834788	Order Number
iccid	String	true	89129858981922456777	ICCID

4.12.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description

Error Code Explanation

subCode	subMsg	Solution
6001	The card type of the current order does not support termination.	Card type does not support termination; cannot terminate this card type.
5003	The order status does not support termination.	Only orders with status 'In Use' or 'Used Up' support termination.
5001	Channel iccid or order mismatches.	Channel ICCID or order number mismatch; check if the order number and card number are consistent for the channel account.
4000	System issue, contact TGT technical support.	System issue, contact TGT technical support.

Request Example 复制代码

```
{
  "iccid":"89852342621027010000",
  "orderNo":"TG202511121917117289"
}
```

Response Example 复制代码

```
{
  "code":"0000",
  "msg":"success"
}
```

4.13 Place top up order

Business Description

Purchase a top-up package for data recharge (Currently only applicable to specific carrier cards (C4 cards) supporting daily packages).

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/order/topup/create

QPS Limit：50

Response Time：500ms

4.13.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
orderNo	String	true	TG2198392592	Order Number
purchaseType	int	true	1	Purchase Type (1: For the current day 2: Daily)
idempotencyKey	String	true	550e8400-e29b-41d4-a716-446655440000	Request ID, maximum length 64 characters. (In exceptional cases like request interruption, timeout, etc., where the channel did not receive a response from the TGT system, pass the same **idempotencyKey** when retrying). Must be a **globally unique** string (UUID v4 is the best choice).

4.13.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description
data	object		Data encapsulation entity, returns a **TopupOrderInfo** object
TopupOrderInfo Object Field Description			
topupNumber	String	TU123141893179414	Top-up Order Number

Error Code Explanation

subCode	subMsg	Solution

5031	The Topup order is currently being processed	Top-up order is currently being processed.
5028	The order does not exist.	Order does not exist.
5018	Order status does not support topup	Order status does not support top-up.
5019	The product type does not support topup	Product type does not support top-up.
5020	The product card type does not support topup	Product card type does not support top-up.
5015	The product package is not available for topup	Product package is not available for top-up (or top-up price is not configured).
5023	The order has expired	Order has expired (not within the valid usage period).
5017	Test coin orders do not allow ordering fuel packs	Test coin orders (and non-manufacturer orders) do not allow ordering top-up packages.

Request Example 复制代码

```
{
  "orderNo": "TG2198392592",
  "purchaseType": 1,
  "idempotencyKey": "550e8400-e29b-41d4-a716-446655440003"
}
```

Response Example 复制代码

```
{
  "code": "0000",
  "msg": "success",
  "subCode": "",
  "subMsg": "",
  "data": {
    "topupNumber": "TU123141893179414"
  }
}
```

4.14 Paginated Query of Top-up Orders

Business Description

Paginated query of all top-up orders under the channel account. Returns all order records by default, sorted by creation time in descending order. Supports querying top-up orders within a specific creation time range.

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/order/topup/list

QPS Limit：50

Response Time：500ms

4.14.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
pageNum	int	true	1	Page number to query (>=1)
pageSize	int	true	10	Number of items per page, maximum 100
createdStartTime	String	false	2025-07-10T00:00:00Z	Start time for querying order creation time, format: yyyy-MM-dd'T'HH:mm:ss'Z'
createdEndTime	String	false	2025-07-10T23:59:59Z	End time for querying order creation time, format: yyyy-MM-dd'T'HH:mm:ss'Z'
lang	String	false	en	Language (English: en, Japanese: ja-JP, Chinese: zh-CN), default is English

4.14.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes

subCode	String		Business Error Code
subMsg	String		Business Error Description
data	object		Data encapsulation entity, returns a **TopupOrderRes** object
TopupOrderRes Object Field Description			
total	Integer	1	Total number of records
list	JSONArray		Data encapsulation entity, returns an array of **TopupOrderInfo** objects
TopupOrderInfo Object Field Description			
iccid	String	20250709181134000023	ICCID
orderNo	String	TG202507111329423579	Order Number
topupNumber	String	TU2507111752212546664001	Top-up Order Number
productName	String	CT-C4-addon-test-esim-plus-1	Package Name Supports internationalization; automatically matches and returns the product name in the language specified by the 'lang' parameter.
productType	String	DAILY_PACK	Product Type
topupName	String	100MB add-on	Top-up Package Name
quantity	String	1	Quantity of top-up packages ordered
channelStartTime	String	2025-07-10T18:30:14Z	Top-up package start time (UTC+0)
channelEndTime	String	2025-07-11T18:30:14Z	Top-up package end time (UTC+0)
settlementCurrency	String	CNY	Settlement currency
settlementAmount	Decimal(28, 2)	111.00	Settlement amount (2 decimal places)

channelCreatedTime	String	2025-07-11T05:42:27Z	Creation time (UTC+0)

Request Example 复制代码

```
{
  "pageNum": 1,
  "pageSize": 10,
  "createdStartTime": "2025-07-10T00:00:00Z",
  "createdEndTime": "2025-07-10T23:59:59Z",
  "lang": "en"
}
```

Response Example 复制代码

```
{
  "code": "0000",
  "msg": "success",
  "subCode": "",
  "subMsg": "",
  "data": {
    "total": 3,
    "data": [
      {
        "iccid": "20250709181134000023",
        "orderNo": "TG202507111329423579",
        "topupNumber": "TU25071117522125466640 01",
        "productName": "CT-C4-addon-test-esim-plus-1",
        "productType": "DAILY_PACK",
        "topupName": "100MB add-on",
        "quantity": "1",
        "channelStartTime": "2025-07-10T18:30:14Z",
        "channelEndTime": "2025-07-11T18:30:14Z",
        "settlementCurrency": "CNY",
        "settlementAmount": 111.00,
        "channelCreatedTime": "2025-07-11T05:42:27Z"
      }]}}
```

4.15 Query Top-up Orders Associated with the orderNo

Business Description

Query the top-up orders associated with an order number.

Interface Description

Request URL：BaseUrl+/eSIMApi/v2/order/topups

QPS Limit：50

Response Time：500ms

4.15.1 Request Parameters

Parameter Name	Type	Required	Example Value	Description
pageNum	int	true	1	Page number to query (>=1)
pageSize	int	true	10	Number of items per page, maximum 100
orderNo	String	false	20250709181134000023	Order Number (At least one of orderNo, iccid, or topupNumber must be provided)
iccid	String	false	TG202507111329423579	ICCID (At least one of orderNo, iccid, or topupNumber must be provided)
topupNumber	String	false	TU2507111752212546664001	Top-up Order Number (At least one of orderNo, iccid, or topupNumber must be provided)
lang	String	false	en	Language (English: en, Japanese: ja-JP, Chinese: zh-CN), default is English

4.15.2 Response Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
subCode	String		Business Error Code
subMsg	String		Business Error Description

data	JSONArray		Data encapsulation entity, returns a **TopupOrderRes** object
TopupOrderRes Object Field Description			
total	Integer	1	Total number of records
list	JSONArray		Data encapsulation entity, returns an array of **TopupOrderInfo** objects
TopupOrderInfo Object Field Description			
iccid	String	20250709181134000023	ICCID
orderNo	String	TG202507111329423579	Order Number
topupNumber	String	TU2507111752212546664001	Top-up Order Number
productName	String	CT-C4-addon-test-esim-plus-1	Package Name Supports internationalization; automatically matches and returns the product name in the language specified by the 'lang' parameter.
productType	String	DAILY_PACK	Product Type
topupName	String	100MB add-on	Top-up Package Name
quantity	String	1	Quantity of top-up packages ordered
channelStartTime	String	2025-07-10T18:30:14Z	Top-up package start time (UTC+0)
channelEndTime	String	2025-07-11T18:30:14Z	Top-up package end time (UTC+0)
settlementCurrency	String	CNY	Settlement currency
settlementAmount	Decimal(28,2)	111.00	Settlement amount (2 decimal places)

channelCreatedTime	String	2025-07-11T05:42:27Z	Creation time (UTC+0)

Error Code Explanation

subCode	subMsg	Solution
5030	iccid, topupNumber, orderNo, at least one is not empty!	At least one of iccid, topupNumber, or orderNo must be provided.

Request Example 复制代码

```
{
  "pageNum": 1,
  "pageSize": 10,
  "orderNo": "TG20250926180107141009 2",
  "iccid": "8982052207013909758F",
  "topupNumber": "TU25071117522125466640 01",
  "lang": "en"
}
```

Response Example 复制代码

```
{
  "code": "0000",
  "msg": "success",
  "data": {
    "total": 2,
    "data": [
      {
        "iccid": "8982052207013909758F",
        "orderNo": "TG20250926180107141009 2",
        "topupNumber": "TU25071117522125466640 01",
        "productName": "CT-C4-addon-test-esim-plus-1",
        "productType": "DAILY_PACK",
        "topupName": "1GB add-on",
        "quantity": "1",
        "channelStartTime": "2025-09-27T00:00:00Z",
        "channelEndTime": "2025-09-27T23:59:59Z",
        "settlementCurrency": "CNY",
        "settlementAmount": 2.50,
        "channelCreatedTime": "2025-09-26T18:05:00Z"
```

```
      },
      {
        "iccid": "8982052207013909758F",
        "orderNo": "TG202509261801071410092",
        "topupNumber": "TU2507111752212546664002",
        "productName": "CT-C4-addon-test-esim-plus-1",
        "productType": "DAILY_PACK",
        "topupName": "500MB add-on",
        "quantity": "1",
        "channelStartTime": "2025-09-28T00:00:00Z",
        "channelEndTime": "2025-09-28T23:59:59Z",
        "settlementCurrency": "CNY",
        "settlementAmount": 1.00,
        "channelCreatedTime": "2025-09-27T10:30:00Z"
      }
    ]
  }
}
```


5 Webhook notification

General Callback Interaction Rules

Define the webhook callback interaction standards between the TGT system and the channel system, covering response format requirements to avoid callback failures due to non-standard interactions.

Channel Response Specification (Mandatory)

After the TGT system sends a callback request, the channel must return a response in the specified format; otherwise, it will be judged as a callback failure and trigger a retry.

- Response Format: Only `application/json` type is supported. Fields must strictly follow camelCase format without additional nesting.
- Required Fields:

Field Name	Type	Value Requirement	Description
code	String	Fixed as "0000"	Indicates that the channel has successfully received and processed the callback (regardless of business logic success, only confirmation of "receipt" is needed).

msg	String	Fixed as "success"	Response description, only this fixed value is supported.

Error Response: If a business exception occurs when the channel receives the callback (e.g., order does not exist), **the channel must still return the successful response above first**, and then handle the business exception internally (must not return a response code other than "0000", otherwise TGT will retry).

Response Example 复制代码

```
{
  "code": "0000",
  "msg": "success"
}
```

- **Timeout Requirement:** The channel must return a response within **10 seconds**. Failure to respond within **10 seconds** will be judged by TGT as a "callback timeout".
- **Retry Mechanism:** Retries occur every 5 seconds for up to 2 hours, after which retries stop.

Callback Signature Verification

- **Core Signature Rules**

To ensure data transmission security and prevent parameters from being tampered with during transmission, all requests must carry a digital signature (sign) calculated from the parameters. After receiving the request, the server will recalculate the signature using the same rules and compare it. Only requests with matching signatures will be processed.

Signature Generation Steps:

1. **Filter Valid Parameters**

   Ignore the following two types of parameters; they do not participate in the signature calculation:
   - The `sign` parameter itself, which is included in the request.
   - Any binary file parameters (such as `byte[]` type).
2. **Sort Parameters**

   Sort all remaining parameters alphabetically by parameter name based on ASCII code, in ascending order (i.e., A-Z, 0-9).
3. **Concatenate String**

   Directly concatenate the sorted parameters in the format `parameter name + parameter value`, without adding any symbols or delimiters in between.

   Example: Parameters `name=John` and `age=25` would be concatenated as `nameJohnage25`.
4. **Add Security Secret Key**

   Add the exclusive secret key (`secret`) assigned to you to both the beginning and the end of the concatenated string.

   Format: `secret` + `concatenated parameter string` + `secret`

5. **Generate Signature**
   Calculate the **MD5** hash of the final string. The resulting hash value is the signature (`sign`) for this request.

(Note: The `secret` must be kept strictly confidential. It is recommended to add a `timestamp` parameter to set a validity period for the signature to enhance security.)

- **Key Example**

Original parameters: foo:1, bar:2, foo_bar:3, foobar:4, sign:xxx, file:byte []

After filtering and sorting: bar:2, foo:1, foo_bar:3, foobar:4

Parameter concatenation result: bar2foo1foo_bar3foobar4

Signature calculation formula: sign = MD5(secret + "bar2foo1foo_bar3foobar4" + secret)

- **Signature Algorithm (Java, PHP, Python examples provided in original document - omitted here for brevity in translation, but essential logic described above)**

JAVA 复制代码

```java
    private static final String MD5 = "MD5";
    public static final String SIGN = "sign";

    public static String getStringToSign(JSONObject params, String secretKey) {
        //Filter and flatten all fields (exclude sign and empty values, handle
nesting)
        List<String> keyValueList = new ArrayList<>();
        flattenParams(params, "", keyValueList);

        // Sort by field name ASCII ascending order
        Collections.sort(keyValueList);

        // Concatenate all key+value (no separators)
        StringBuilder keyValueStr = new StringBuilder();
        for (String kv : keyValueList) {
            keyValueStr.append(kv);
        }

        // Concatenate secret + key-value string + secret
        String signSource = secretKey + keyValueStr + secretKey;

        return signSource;
    }

    // Recursively flatten nested Map: convert key=value to keyvalue string
and store in list
```

```java
    private static void flattenParams(Map<String, Object> params, String
parentKey, List<String> keyValueList) {
        for (Map.Entry<String, Object> entry : params.entrySet()) {
            String key = entry.getKey();
            Object value = entry.getValue();

            // Exclude sign field
            if (SIGN.equals(key)) {
                continue;
            }

            // Exclude empty values (null, empty strings)
            if (value == null || value.toString().trim().isEmpty()) {
                continue;
            }

            // Handle nested objects (recursive flattening)
            String currentKey = parentKey.isEmpty() ? key : parentKey + "." +
key;
            if (value instanceof Map) {
                flattenParams((Map<String, Object>) value, currentKey,
keyValueList);
            } else {
                // Non-object type: concatenate key+value (convert value to
string)
                keyValueList.add(currentKey + value);
            }
        }
    }

    public static String getMd5Str(String str) {
        return toHex(getMd5(str));
    }

    private static byte[] getMd5(String str) {
        MessageDigest messageDigest;
        try {
            messageDigest = MessageDigest.getInstance(MD5);
            messageDigest.reset();
            messageDigest.update(str.getBytes(StandardCharsets.UTF_8));
        } catch (NoSuchAlgorithmException e) {
            log.error("NoSuchAlgorithmException caught!", e);
            return ArrayUtils.EMPTY_BYTE_ARRAY;
        }
        return messageDigest.digest();
```

```
    }

    private static String toHex(byte[] byteArray) {
        if (null == byteArray) {
            return null;
        }
        StringBuilder md5Str = new StringBuilder();
        for (byte b: byteArray) {
            md5Str.append(String.format("%02x", b));
        }
        return md5Str.toString();
    }
```

PHP 复制代码

```
  public static function sign($params, $secret) {
        // 1. Filter and flatten all fields (exclude sign and empty values,
handle nesting)
        $keyValueList = [];
        self::flattenParams($params, '', $keyValueList);

        // 2. Sort by field name ASCII ascending order
        sort($keyValueList, SORT_STRING);

        // 3. Concatenate all key+value (no separators)
        $keyValueStr = implode('', $keyValueList);

        // 4. Concatenate secret + key-value string + secret
        $signSource = $secret . $keyValueStr . $secret;

        // 5. MD5 signature (32-bit lowercase)
        return md5($signSource);
    }

    // Recursively flatten nested arrays: convert key=value to keyvalue string
and store in array
    private static function flattenParams($params, $parentKey, &$keyValueList)
{
        foreach ($params as $key => $value) {
            // Exclude sign field
            if ($key === 'sign') {
                continue;
            }

            // Exclude empty values (null, empty strings)
```

```
            if ($value === null || trim((string)$value) === '') {
                continue;
            }

            // Handle nested arrays (recursive flattening)
            $currentKey = $parentKey === '' ? $key : $parentKey . '.' . $key;
            if (is_array($value)) {
                self::flattenParams($value, $currentKey, $keyValueList);
            } else {
                // Non-array type: concatenate key+value (convert value to
string)
                $keyValueList[] = $currentKey . (string)$value;
            }
        }
    }
```

Python 复制代码

```
    @staticmethod
    def sign(params: Dict[str, Any], secret: str) -> str:
        # 1. Filter and flatten all fields (exclude sign and empty values,
handle nesting)
        key_value_list: List[str] = []
        SignUtil._flatten_params(params, '', key_value_list)

        # 2. Sort by field name ASCII ascending order
        key_value_list.sort()

        # 3. Concatenate all key+value (no separators)
        key_value_str = ''.join(key_value_list)

        # 4. Concatenate secret + key-value string + secret
        sign_source = f"{secret}{key_value_str}{secret}"

        # 5. MD5 signature
        md5 = hashlib.md5()
        md5.update(sign_source.encode('utf-8'))
        return md5.hexdigest()

    # Recursively flatten nested dictionary: convert key=value to keyvalue
string and store in list
    @staticmethod
    def _flatten_params(params: Dict[str, Any], parent_key: str,
key_value_list: List[str]) -> None:
        for key, value in params.items():
            # Exclude sign field
```

```
            if key == 'sign':
                continue

            # Exclude empty values (None, empty strings)
            if value is None or str(value).strip() == '':
                continue

            # Handle nested dictionaries (recursive flattening)
            current_key = f"{parent_key}.{key}" if parent_key else key
            if isinstance(value, dict):
                SignUtil._flatten_params(value, current_key, key_value_list)
            else:
                # Non-dictionary type: concatenate key+value (convert value to
string)
                key_value_list.append(f"{current_key}{str(value)}")
```

5.1 eSim Callback

5.1.1 Place Order Webhook (Standard eSIM)

Business Description

- Trigger: Triggered only when the card issuance process for a "Create Order" request (regular eSIM / push mode eSIM) is completed (regardless of success or failure).
- After the card issuance task is submitted, the asynchronous processing result will be sent to the channel's provided card issuance notification callback URL. This section describes the parameter format sent by the TGT system to the channel. The channel needs to develop an interface to receive this notification. Upon receiving TGT's callback notification, the channel must respond to the TGT system. The response format must be **{"code":"0000","msg":"success"}.**

(Note: This is not a channel API integration interface. If the channel's customer system has whitelist restrictions, it must be informed to add our system IP addresses.)

5.1.1.1 Callback Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
timestamp	String	2025-08-01T00:00:00Z	Signature timestamp (UTC+0)

sign	String	qweasdzxcdsaq	Signature parameter
data	Object		Data encapsulation entity, returns a **CallbackInfo** object
CallbackInfo Object Field Description			
eventType	int	1	Notification event enumeration: 1: Card Issuance Notification
businessType	String	sucess	Business type, for ESIM business: ESIM
idempotencyKey	String	550e8400-e29b-41d4-a716-446655440000	Returned if the channel's initial request included it; otherwise, this parameter is absent.
orderInfo	Object		Data encapsulation entity, returns an **OrderInfo** object
OrderInfo Object Field Description			
orderNo	String	SE20251110111104020001	Order Number
iccid	String	89972502300001111194F	ICCID
qrCode	String	LPA:1$esiminfra.toprsp.com$7ADBC7BBD32B47438D0BB3F411111DB6	eSIM profile download address. The channel can use the content of this field to generate a QR code. The format is: LPA:1$∗∗∗$∗∗∗
channelOrderNo	String		Channel Order Number, defined by the channel
imsi	String	260036615121111	IMSI
msisdn	String	837383282	MSISDN, phone number
rentalContractNumber	String	8182551111	Rental contract number (returned only for P1, otherwise not returned)
activatedStartTime	String	2025-11-10T14:26:04Z	Activation/Usage start time (returned only for specified activation)

activatedEndTime	String	2026-12-09T14:26:03Z	Activation/Usage end time (returned only for specified activation)
latestActivationTime	String	2026-01-09T14:26:03Z	Latest activation time (returned only for auto-activation)
renewExpirationTime	String	2026-01-16T14:26:03Z	Latest renewal time (not returned for orders that do not support renewal)
createdTime	String	2025-11-10T14:26:04Z	Order creation time
orderType	String	MULTIPLEMONTHS_AUTO	Order Type

Error Code Explanation

code	msg	Solution
5044	The order failed, please contact customer service or administrator	Order creation failure, e.g., no available cards, etc.
4012	The order start date cannot be less than the current time	Order start date cannot be earlier than the current time.
4007	The product belongs to the incorrect platform	Product platform is incorrect.
4009	This channel is not configured for the product price	Product price organization status issue; contact TGT to check if the price is configured for this product.
4011	Package settlement price is illegal!	Product package configuration issue.
4013	product is unvisible,can not to add order	Product is not listed/shelf; list the product. Or, this device does not support purchasing this product.
4008	The channels of this business model do not support calling this interface	Account attributes do not match.
4016	The purchase of producttype of data package is not supported 流	Data packages only support regular data packs, not other forms.

4010	Channel account balance is insufficient, please top up	Channel account balance is insufficient; please recharge.
4000	Failure, please contact the service staff!	System issue, contact TGT technical support.
4004	ESIM products only support ESIM device model group products	ESIM interface only supports ordering products from ESIM device model groups.

5.1.2 Renewal webhook (Standard eSIM)

Business Description

When a renewal task is submitted, the asynchronous processing result will be sent to the channel's provided notification callback address. This section describes the parameter format sent by the TGT system to the channel. The channel needs to develop an interface to receive this notification. Upon receiving TGT's callback notification, the channel must respond to the TGT system. The response format must be **{"code":"0000","msg":"success"}**

(Note: This is not a channel API integration interface. If the channel's customer system has whitelist restrictions, it must be informed to add our system IP addresses.)

5.1.2.1 Callback Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
timestamp	String	2025-08-01T00:00:00Z	Signature timestamp (UTC+0)
sign	String	qweasdzxcdsaq	Signature parameter
data	Object		Data encapsulation entity, returns a **CallbackInfo** object
CallbackInfo Object Field Description			
eventType	int	2	2: Renewal Notification

businessType	String	ESIM	Business type, for ESIM business: ESIM
idempotencyKey	String	550e8400-e29b-41d4-a716-446655440000	Returned if the channel's initial request included it; otherwise, this parameter is absent.
orderInfo	JSONArray		Data encapsulation entity, returns an array of **OrderInfo** objects
OrderInfo Object Field Description			
orderNo	String	SE20251110111104020001	Order Number
iccid	String	89972502300001111194F	ICCID
qrCode	String	LPA:1$esiminfra.toprsp.com$7ADBC7BBD32B47438D0BB3F411111DB6	eSIM profile download address. This field is returned only when `operateType` is 1 and `code` is 0000. The channel can use it to generate a QR code. Format: LPA:1$***$***
channelOrderNo	String	12345	Channel Order Number, defined by the channel
imsi	String	260036615121111	IMSI
msisdn	String	837383282	MSISDN, phone number
activatedStartTime	String	2025-11-10T14:26:04Z	Activation/Usage start time (returned only for specified activation)
activatedEndTime	String	2026-12-09T14:26:03Z	Activation/Usage end time (returned only for specified activation)
latestActivationTime	String	2026-01-09T14:26:03Z	Latest activation time (returned only for auto-activation)
renewExpirationTime	String	2026-01-16T14:26:03Z	Latest renewal time (not returned for orders that do not support renewal)
createdTime	String	2025-11-10T14:26:04T	Order creation time

orderType	String	MULTIPLEMONTHS_AUTO	Order Type

Error Code Explanation

code	msg	Solution
4010	Channel account balance is insufficient, please top up	Channel account balance is insufficient; please recharge.
4000	Failure, please contact the service staff!	System issue, contact TGT technical support.
4004	ESIM products only support ESIM device model group products	ESIM interface only supports products from ESIM device model groups.

5.1.3 Place Order Webhook（SMS Push eSIM）

Business Description

When a push mode task is submitted, the asynchronous processing result will be sent to the channel's provided notification callback address. This section describes the parameter format sent by the TGT system to the channel. The channel needs to develop an interface to receive this notification. Upon receiving TGT's callback notification, the channel must respond to the TGT system. The response format must be **{"code":"0000","msg":"success"}**

(Note: This is not a channel API integration interface. If the channel's customer system has whitelist restrictions, it must be informed to add our system IP addresses.)

5.1.3.1 Callback Parameters

Parameter Name	Type	Example Value	Description
code	String	0000	0000=Success
msg	String	sucess	Information about error codes
timestamp	String	2025-08-01T00:00:00Z	Signature timestamp (UTC+0)
sign	String	qweasdzxcdsaq	Signature parameter

data	Object		Data encapsulation entity, returns a **CallbackInfo** object
CallbackInfo Object Field Description			
eventType	int	3	3: Callback Card Issuance Notification
businessType	String	ESIM	Business type, for ESIM business: ESIM
idempotencyKey	String	550e8400-e29b-41d4-a716-446655440000	Returned if the channel's initial request included it; otherwise, this parameter is absent.
orderInfo	Object		Data encapsulation entity, returns an **OrderInfo** object
OrderInfo Object Field Description			
orderNo	String	SE20251010184131920001	Order Number
channelOrderNo	String	2222	Channel Order Number, defined by the channel
activatedStartTime	String	2025-10-20T04:59:59Z	Activation/Usage start time (returned only for specified activation)
activatedEndTime	String	2025-10-20T04:59:59Z	Activation/Usage end time (returned only for specified activation)
createdTime	String	2025-10-10T23:41:32Z	Order creation time
orderType	String	DAYPASS	Order Type

Error Code Explanation

code	msg	Solution

4007	The product belongs to the incorrect platform	Check product package configuration.
4009	This channel is not configured for the product price	Product price organization status issue.
4011	Package settlement price is illegal!	Product package configuration issue.
4013	product is unvisible,can not to add order	Product is not listed/shelf; list the product.
4008	The channels of this business model do not support calling this interface	Account attributes do not match.
4010	Channel account balance is insufficient, please top up	Channel account balance is insufficient; please recharge.
4000	Failure, please contact the service staff!	System issue, contact TGT technical support.

6 Error Codes

6.1 Public Error Codes

code	msg	Solution
1003	[{Parameter key}] can't be blank [{Parameter key}] can't be null [{Parameter key}] can't be empty [{Parameter key}] length must be less than {max} [{Parameter key}] must be {param1}, {param2}... or {param3} [{Parameter key}] style must be yyyy-MM-dd [{Parameter key}] style must be yyyy-MM-dd'T'HH:mm:ss'Z' [{Parameter key}] Parameter types do not match!	Check parameters: parameter is empty, wrong format, illegal value, out of bounds, etc.

1004	page parameter error	Pagination parameter exception. Check parameters: pageNum must be >=1, pageSize not greater than 100.
1005	Character set error, please check	Character set error, request parameter charset is wrong. Currently supported format: UTF-8.
2000	The service is unavailable, please try again later	Service unavailable, try again later.
2001	Insufficient interface permission	Insufficient permissions. The current account lacks interface permissions. Please contact TGT administrator for confirmation.
2003	Token invalid	Token invalid. Obtain a new token via the **Authorization to get new token** interface.
2004	Token unknown	Token unknown. Obtain a new token via the **Authorization to get new token** interface.
0429	Frequent requests, please try again later	Access restricted. QPS limit exceeded. Requests beyond the threshold are rejected. Requests within the limit will return normally.
5000	Remote service error	Business processing failed. Refer to the specific interface's business error code.

7 FAQ

Q1: How do I obtain the Account ID and Secret Key parameters?

A1: Both the Account ID and Secret Key are required for production and sandbox environments and are provided by Tuge Technology. You must complete API testing in the sandbox environment before switching to production.

Q2: What is the difference between the Sandbox and Production environments?

A2: The Sandbox environment is for API testing during the integration phase. You can test all API endpoints here without incurring actual order charges. eSIMs in the Sandbox cannot be activated on physical devices. The Production environment is for commercial use. Your account must be pre-funded, and all orders will generate actual charges.

Q3: How can I verify if an API test order is successful in the Sandbox?

A3: After placing an order via API, you can log in to the Sandbox portal at https://enterpriseapisandbox.tugegroup.com to review the order details.

Q4: What is the difference between Auto-Activate and Activate-on-Order?

A4:AUTO_ACTIVATE: The eSIM can be activated on any day within the order's validity period. Once activated, the order's validity period is refreshed to become the actual usage period. For example, an 18GB/15-day plan (60-day validity) can be activated anytime within 60 days. The 15-day usage period starts from the activation date.

ACTIVATE_ON_ORDER: The activation start date is specified at the time of order creation. The eSIM cannot be used before this date. The usage period is fixed upon order creation and does not refresh.

Q5: What does "Cycle Type Timezone" mean?

A5: The cycle type (24-hour or calendar day) is determined by carrier rules and varies by eSIM.

24-hour cycle: The usage period is based on 24-hour increments. For example, a 1-day plan activated at 2025-11-22 13:33:47 will expire at 2025-11-23 13:33:46.

Calendar day cycle: The usage period follows the carrier's designated timezone (e.g., UTC+8). A 1-day plan activated at 2025-11-25 13:48:35 may expire at 2025-11-26 08:00:00.

Q6. How often is the product packages updated? If there are updates, how can I obtain them?

A6: The product plan does not have a fixed update schedule. You can use the interface in section 4.1 to query all product information, store it in your database, and periodically retrieve the latest details. Additionally, when product packages change, the business team will also synchronize updates to the plan.

Q7: Is a callback interface mandatory?

A7: Yes. Since the order creation process is asynchronous, a callback interface is required to receive order status notifications (e.g., success/failure). Without it, you would need to repeatedly poll the "Paginated Order Query" API (section 4.7), which is not the recommended integration approach.

Q8: How is the eSIM QR code generated?

A8: After order creation, eSIM details (including the qrCode field) are sent to your callback interface. You can convert this qrCode data into a scannable QR code using your own code.

Q9: Can I customize the product package name?

A9: Yes. Orders are created using the unique productCode value. You may display any custom name for the product package in your system as long as it correctly maps to the corresponding productCode.

Q10: What is the difference between a package Renewal and a Data Add-on?

A10:Package Renewal: Allows repurchasing a data plan (same or different) for the same eSIM, extending both the data allowance and the validity period. Supported by multiple eSIM types.

Place top up order: Only available for C4-type daily plans. It adds extra data for one or more days to the existing plan without extending the eSIM's overall validity period. Ideal for daily plans that have exhausted high-speed data and entered throttled speeds.

Q11: What product packages are available in the Sandbox environment for API testing?

A11: You can retrieve all available product packages for API testing in the Sandbox environment by calling the "Paginated Query All eSIM Product packages" API. Alternatively, refer to **Appendix 1** for the product plan list.

Q12: How can I categorize product packages by country/region?

A12: The "Paginated Query All eSIM Product packages" API response includes the countryCodeList (country codes) and mccList (Mobile Country Code collections) fields. You can use these parameters to classify product packages by geographic region. For detailed information on countryCodeList and mccList, please see **Appendix 2.**

Q13: How to use Universal Links for direct eSIM installation on iOS?

A13: With the introduction of Universal Links by iOS, users running iOS 17.4 or later can install eSIMs directly via a dedicated URL. If your customers are using iOS 17.4+, you can provide them with this URL. Implementation: After order creation, you will receive a qrCode value. Concatenate the official Apple link https://esimsetup.apple.com/esim_qrcode_provisioning?carddata= with the qrCode value to generate the one-tap installation URL for iOS.Example URL: https://esimsetup.apple.com/esim_qrcode_provisioning?carddata=1$esiminfra.toprsp.com$D0785A57B85C46858A96F802F8E811A1

8 Release Version

Version	Release Time	Author	Description
V2.0	20251124	Yujia.tang	Initial Version

9 Appendix

Appendix 1

Product Code	Product Name	Product Type	Card Type Coding

A-002-ES-AU-DZ-A1-8D/120D-17GB	【Esim】AIS Thailand 17GB/8days（AU)(Valid for 120 days)	Data package	A1
A-166-ES-AU-DZ-A1-8D/60D-6GB	【Esim】AIS Asia10 6GB/8days（AU)(Valid for 60 days)	Data package	A1
E-184-ES-ZD-eO1-D-10D/60D-10GB	【Esim】Europe (11 Countries) 10 Days (10GB High-Speed Data/Day) (eO1) (ZD)	Daypass	Euro-eO1
A-002-ES-AU-T-30D/180D-3GB(A)	【ESIM】Israel 3GB/30 Days (180-Day Validity) (M1) (A)	Data package	M1
A-013-ES-AU-eO1-T-3D/60D-2GB	【Esim】North America (3) 2GB/3 days (valid for 60 days) (eO1) (AU)	Data package	Euro-eO1
A-136-ES-AU-C4-1D/60D-1GB	【Esim】Asia 5 Countries 1 Day (1GB High-Speed/Day)(C4)(AU)	Daypass	C4
A-167-ES-AU-D-A1-8D/60D-U(A)	【Esim】AIS Asia 8 Days (Unlimited/Day) (60-Day Validity) (A1)	Daypass	A1
A-002-ES-AU-D-30D/180D-500MB	【ESIM】Israel 500MB/30 days (valid for 180 days) (M1) (A1)	Daypass	M1
E-02-ES-eP1-ZD-T-30D/60D-5GB(A)	【ESIM】Russia 5GB/30 days (eP1)	Data package	Euro-ep1
A-002-ES-ZD-C4-6D/60D-500MB	【Esim】Japan 6 Days (500MB High-Speed Data/Day) (C4)	Data package	C4
E-01-F2-ES-AU-T-7D/60D-1GB(A)	【ESIM】UK 1GB/7 days (valid for 60 days) (F2)	Data package	F2
E-02-F2-ES-ZD-T-7D/60D-1GB(A)	【ESIM】UK 1GB/7 days (valid for 60 days) (F2)	Data package	F2
A-001-ES-AU-C4-5D/60D-1GB	【Esim】Japan 5 Days (500MB High-Speed Data/Day) (C4)	Data package	C4
E-01-ES-eP1-AU-T-15D/60D-3GB(A)	【ESIM】Austria 3GB/15 Days (60-Day Validity) (eP1)	Data package	Euro-ep1

B-01-ES-AU-eP1-D-1D/60D-500MB(A)	【Esim】US & Canada 1-Day Plan (500MB High-Speed Data/Day)(eP1)(AU)	Daypass	Euro-ep1
B-02-ES-ZD-eP1-D-3D/60D-500MB(A)	【Esim】US & Canada 3 Days (500MB High-Speed Data/Day)(eP1)(AU)	Daypass	Euro-ep1
A-136-ES-ZD-C4-1D/60D-1GB	【Esim】Asia 5 Countries 1 Day (1GB High-Speed/Day)(C4)(ZD)	Daypass	C4
A-013-ES-ZD-eO1-T-3D/60D-2GB	【Esim】North America (3) 2GB/3 days (valid for 60 days) (eO1) (ZD)	Data package	Euro-eO1
E-184-ES-AU-eO1-D-10D/60D-10GB	【Esim】Europe (11 Countries) 10 Days (10GB High-Speed Data/Day) (eO1) (AU)	Daypass	Euro-eO1

Appendix 2

Country Code	Country name	mcc
GE	Georgia	282
PM	Saint Pierre and Miquelon	308
CL	Chile	730
CH	Switzerland	228
ZM	Zambia	645
GW	Guinea-Bissau	632
IQ	Iraq	418
CD	Democratic Republic of Congo (DRC)	630
KR	Korea	450
CM	Cameroon	624
LB	Lebanon	415
AA	Global	000

BO	Bolivia	736
TO	Tonga	539
MT	Malta	278
AL	Albania	276
PF	French Polynesia	547
PL	Poland	260
GT	Guatemala	704
KZ	Kazakhstan	401
MA	Morocco	604
AU	Australia	505
ES	Spain	214
GXD	Guernsey	591
LC	Saint Lucia	358
WS	Samoa	549
PK	Pakistan	410
SN	Senegal	608
EE	Estonia	248
HR	Croatia	219
DE	Germany	262
BS	Bahamas	364
DM	Dominica	366
QA	Qatar	427

JO	Jordan	416
MY	Malaysia	502
EG	Egypt	602
GB	UK	234,235
AR	Argentina	722
KN	Saint Kitts and Nevis	356
MR	Mauritania	609
CF	Central Africa	623
SBTD	St Barts	592
ET	Ethiopia	636
GN	Guinea	611
KI	Kiribati	545
CG	Republic of Congo(Brazzaville)	629
UA	Ukraine	255
BJ	Benin	616
BI	Burundi	642
SD	Sudan	634
LS	Lesotho	651
ID	Indonesia	510
PY	Paraguay	744
FM	Micronesia	550
RW	Rwanda	635

TG	Togo	615
TW	Taiwan(China)	466
SY	Syria	417
AG	Antigua and Barbuda	344
GP	Guadeloupe (French Caribbean Region)	340
BD	Bangladesh	470
LU	Luxembourg	270
SM	San Marino	292
AW	Aruba	363
MDD	Isle of Man	584
ER	Eritrea	657
LK	Sri Lanka	413
SZ	Swaziland	653
HU	Hungary	216
SB	Solomon Islands	540
AM	Armenia	283
IN	India	404,405,406
MBLT	Gibraltar	266
US	USA	310,311,312,313,314,315,316
DJ	Djibouti	638
BLW	Bulavan	596
LI	Liechtenstein	295

LY	Libya	606
TM	Turkmenistan	438
GQ	Equatorial Guinea	627
ST	Sao Tome and Principe	626
GD	Grenada	888
TN	Tunisia	605
ZW	Zimbabwe	648
UG	Uganda	641
LV	Latvia	247
GA	Gabon	628
NG	Nigeria	621
MP	Northern Mariana	534
BT	Bhutan	402
VE	Venezuela	734
FO	The Faroe Islands (Denmark)	288
SL	Sierra Leone	619
SG	Singapore	525
BA	Bosnia and herzegovina	218
FJ	Fiji	542
KE	Kenya	639
SK	Slovakia	231
AT	Austria	232

ML	Mali	610
MM	Myanmar	414
BW	Botswana	652
KGZ	Kyrgyzstan	437
UZ	Uzbekistan	434
YE	Yemen	421
MZ	Mozambique	643
NR	Nauru	536
JM	Jamaica	338,352
GDLP	Guadalupe	590
UY	Uruguay	748
AE	United Arab emirates	424,430,431
KY	Cayman Islands	346
NC	New Caledonia	546
FI	Finland	244
GM	Gambia	607
IT	Italy	222
SE	Sweden	240
GR	Greece	202
NI	Nicaragua	710
BY	Belarus	257
CY	Cyprus	280

TC	Turks and Caicos Islands (UK)	376
VN	Vietnam	452
IL	Israel	425
BR	Brazil	724
VG	British virgin islands	348
GL	Greenland	290
ZA	South Africa	655
SGBD	Saipan	595
BE	Belgium	206
ZXD	Jersey	587
EC	Ecuador	740
SI	Slovenia	293
MC	Monaco	212
TZ	Tanzania	640
MV	Maldives	472
CZ	Czech	230
HT	Haiti	372
TJ	Tajikistan	436
CV	Cape Verde	625
KM	Comoros	654
PT	Portugal	268
PH	Philippines	515

DO	Dominican	370
TD	Chad	622
GH	Ghana	620
CU	Cuba	368
JP	Japan	440,441
GF	French Guiana	742
AD	Andorra	213
DZ	Algeria	603
BM	Bermuda	350
AF	Afghanistan	412
GY	Guyana	738
CW	Curaçao (The Netherlands)	362
DK	Denmark	238
VU	Vanuatu	541
KH	Cambodia	456
VC	Saint Vincent and the Grenadines	360
VI	United States Virgin Islands	332
IS	Iceland	274
PR	Puerto Rico	330
BH	Bahrain	426
BB	Barbados	342
PE	Peru	716

MD	Moldova	259
MX	Mexico	334
SV	Salvador	706
NP	Nepal	429
SR	Suriname	746
AO	Angola	631
RE	Reunion	647
BZ	Belize	702
LA	Laos	457
CI	Cote d'Ivoire	612
NE	Niger	614
LR	Liberia	618
NA	Namibia	649
RO	Romania	226
BF	Burkina Faso	613
MU	Mauritius	617
TR	Turkey	286
IR	Iran	432
RU	Russia	250
BN	Brunei	528
CN	China	460
CO	Colombia	732

MW	Malawi	650
OM	Oman	422
TT	Trinidad and Tobago	374
ME	Montenegro	297
TL	East Timor	514
TH	Thailand	520
RS	Serbia	220
MK	Macedonia	294
MN	Mongolia	428
CR	Costa rica	712
KW	Kuwait	419
KOS	Kosovo	221
NO	Norway	242
PW	Palau	552
PG	Papua New Guinea	537
NL	Netherlands	204
CA	Canada	302
MH	Marshall Islands	551
HK	Hong Kong(China)	454
SO	Somalia	637
FR	France	208
PA	Panama	714

SA	Saudi Arabia	420
VA	Vatican	225
NZ	New Zealand	530
MG	Madagascar	646
AS	American Samoa	544
SC	Seychelles	633
BG	Bulgaria	284
AI	Anguilla	365
IE	Ireland	272
LT	Lithuania	246
KP	North Korea	467
AZ	Azerbaijan	400
GU	Guam	535
HN	Honduras	708
CK	Cook Islands	548
MO	Macau(China)	455